Size- and Position-dependent opacity in Hungarian

Abstract

Selective opacity emerges when a syntactic constituent is transparent only to certain operations. The Williams Cycle generalization (Williams, 2003) correlates structural size and opacity: Larger clauses are opaque to more syntactic operations than smaller clauses. Firstly, this paper demonstrates that Hungarian exhibits Williams Cycle effects extensively. Secondly, I bring novel data to show that the opacity of Hungarian embedded clauses depends on their final position in the matrix clause: Clauses ending up higher are opaque to more operations. Finally, I provide a unified analysis for size- and position dependent opacity by introducing a new constraint on the locality of movement. According to this constraint, movement steps must proceed along well-formed f(unctional)seq(uence)-fragments. In the absence of successive cyclic movement in Hungarian, cross-clausal movement is pre-conditioned by the movement of embedded clauses: If an embedded clause adjoins to an equal-sized matrix projection, the sequence of embedded heads is c-commanded by a sequence of matrix heads in a way that their union is a well-formed fseq-fragment. Consequently, sub-extraction can proceed to the matrix specifiers c-commanding the adjunction site. This derives size dependent opacity because larger clauses can become transparent via adjunction later, i.e. their adjunction site is c-commanded by fewer matrix specifiers. Position-dependent opacity follows because embedded clauses that have been made transparent by their adjunction cannot leave their adjunction-sites, i.e. embedded clauses become transparent at the cost of their mobility.

Keywords: selective opacity, Williams Cycle, Hungarian, locality, remnant movement

1 Introduction

Syntactic opacity or syntactic islandhood was originally proposed in an absolute sense (Ross 1967): a constituent is either transparent to all syntactic operations or opaque to all of them. However, opacity was later found to be occasionally *selective*: a constituent may be opaque to certain syntactic operations but transparent to others (a.o. Williams 1974, 2003, 2013; Keine 2016, 2019, 2020; Abels 2012; Wurmbrand 2001, 2015; Müller 2014a,b; Poole 2022; Meadows 2023).

One of the most well-known examples of selective opacity is the ban on *hyperraising* in English (Chomsky 1973). The English CP is not an island: *wh*-movement, which lands in Spec,CP, can proceed from English CPs, as in (1). In contrast, the raising of subjects, which lands Spec,TP cannot proceed from CPs, as in (2).

(1) $[_{\text{CP}}$What$_1$ does it seem $[_{\text{CP}}$ that John has eaten __$_1]]$?

(2) $^*[_{\text{TP}}$John$_1$ seems $[_{\text{CP}}$ __$_1$ has eaten an apple$]]$

While the contrast between (1) and (2) has received various analyses (a.o. Chomsky 1973; Abels 2008), there is a line of research that correlates the opacity pattern of English finite clauses with their *internal structural size* (a.o. Williams 2003; Keine 2016, 2019, 2020; Poole 2022; Meadows 2023). In particular, the ungrammaticality of (2) fits the Williams Cycle generalization in (3): Since C>T in the *f*(unctional) *seq*(uence), movement from CPs cannot land in Spec,TP. In contrast, no such violation occurs in (1) because *wh-movement* lands in Spec,CP. Since it is the internal structural size of the clause that determines its opacity, (3) can be referred to as *size-dependent opacity.* Williams Cycle effects have been attested in a number of other, genetically diverse languages, for example, German (Müller 2014a,b), Hindi (Keine 2016, 2019, 2020), Swahili (Meadows 2023), Mandarin Chinese (Meadows and Yan 2024), and Georgian (Bondarenko 2024).

(3) **Williams Cycle/Size-dependent opacity:** Movement to Spec,βP cannot proceed across a dominating αP, where α > β in the functional sequence. (Williams 2003; Poole 2022)

Egressy (2023, 2025, to appear) has recently argued that Hungarian also exhibits Williams Cycle effects: The size of a clause —more specifically, the size of its left-periphery— determines what movement types are possible from that clause; clauses with a richer left-periphery are opaque to more movement types. In parallel, the landing height of the various sub-extraction types determines from what clause-sizes they can proceed. For example, since the movement of relative pronouns lands in a higher position than focusing, relative pronouns can be sub-extracted from clause-types that are opaque to focusing.

In the Hungarian literature, it has been compellingly argued that clauses obligatorily move out from their VP-internal position, landing in the left-periphery or the right-periphery (Kenesei 1994; É. Kiss 2002). In this paper, I will argue that clausal movement correlates with subextraction possibilities from the moved clauses, in a way that relates to the WIlliams Cycle effects documented by Egressy (2025, to appear). To illustrate, consider (4) and (5), where the embedded clause is moved to a topic position preceding the universal quantifier *mindenki* 'everyone'. While relativization is grammatical from topic clauses in (4), focusing is not in (5).[1]

(4) *A város,* $[_{REL}$ *amelyik-en*$]_2$ $[_{TOP}$ *hogy keresztül fut-ott-am* $\_\_{}_2]_1$ *mindenki hall-ott-a*$\_\_{}_1$.
the city which-SUP that across run-PST-1SG.INDF everyone hear-PST-3SG.DEF
'The city, which, that we ran across it, everyone heard it.'

(5) *$[_{TOP}$ *Hogy keresztül fut-ott-am* $\_\_{}_2]_1$ *mindenki* $[_{FOC}$ *CSAK A VÁROS-ON*$]_2$ *hall-ott-a* $\_\_{}_1$
that across run-PST-1SG.INDF everyone only the city-SUP hear-PST-3SG.DEF
Intended: 'That I ran across ONLY THE CITY, everyone heard it.'

Relativization and focusing differ in terms of *landing heights*: While relative pronouns land above topics, foci land below topics in the left-periphery (Horvath 1981; Egressy 2025). In terms of timing, this means that the movement of relative pronouns can proceed from topicalized clauses when the relative movement happens *after* the clause has moved to its topic position, whereas focusing cannot proceed from topicalized clauses when focusing would happen

[1] The abbreviations are the following: 1 = first person, 2 = second person, 3 = third person, ACC = accusative, SADV = sentential adverb, COND = conditional, DEF = definite, FOC = focus, IN = inessive, INDF = indefinite, INF = infinitive, INS = instrumental, PART = particle, PL = plural, POT = potential, PST = past, Q = quantifier, REL = relative, SG = singular, SUB = sublative, SUBJ = subjunctive, SUP = superessive, TOP = topic
The uncited Hungarian data comes from my own data collection. Although I am a native speaker of Hungarian, the grammaticality of critical sentences were double-checked in two surveys. The Hungarian data of the paper comes from the standard dialect spoken in Budapest. The first one was conducted in Summer, 2022, the second one in Spring, 2023. The surveys had 15 and 35 native-speaker participants, respectively.

before the clause reaches its topic position. This pattern is generalized as *position-dependent* opacity in (6): Clauses must reach their final position in the left-periphery before sub-extraction can proceed from them. In other words, Hungarian clauses exhibit a ban on remnant movement (Fiengo 1977; Lasnik and Saito 1994), whereas Hungarian clauses are not frozen to sub-extraction after they land in the left-periphery (cf. Müller 2014a).

(6) **Position-dependent opacity (simplified):** Clauses are transparent to sub-extraction to the left-periphery only after they reach their final position in the left-periphery.

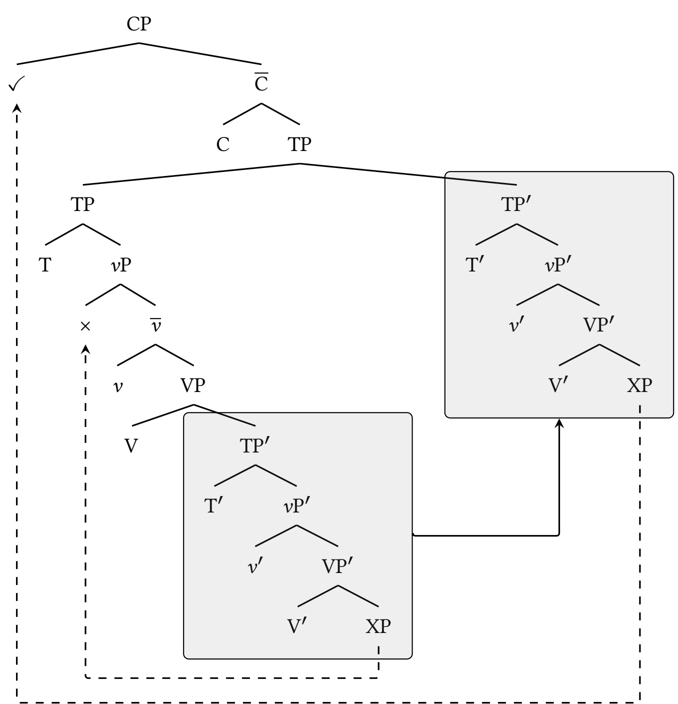
In sum, the opacity of Hungarian clauses depends on two factors: the size/richness of their left-peripheries and their final position in the matrix left-periphery. In this paper, I propose a unified account of size- and position-dependent opacity. In particular, I derive size-dependent opacity from an insight from position-dependent opacity: Embedded clauses become transparent to sub-extraction only after they move out from their VP-internal position and merge with a matrix position. Thus, the first component of my account is that clauses must move to become transparent, to be "unlocked".

However, existing "unlocking proposals" (e.g. Rackowski and Richards 2005; Van Urk and Richards 2015) follow standard Phase Theory (Chomsky 2000, 2001) in that locality domains correspond to certain categories, e.g. CPs are locality domains (phases). Since this rigid, category-specific notion of locality cannot derive selective opacity (Keine 2019), my account follows Bošković (2014) by claiming that syntactic locality should be constrained in terms of *extended projections* and *functional sequences*. Thus, the second component of my analysis is the locality constraint *Move Along Increasing Fseq Indicies* (MAIF) in (7), which is an operationalization of Bošković's intuition. Under MAIF, any clause-internal movement is possible because such movement crosses the heads of a single extended projection. However, cross-clausal movement is more restricted. Using the standard clausal functional sequence V<*v*<T<C in (8), one can see that as long as an embedded TP′ clause is internal to the matrix VP, no XP can be sub-extracted from it because it would cross the V-*v*-T-V-... sequence. This movement violates MAIF because V is not the next head above T in the functional sequence. Therefore, no movement can leave the *in-situ* embedded TP and land in matrix positions like Spec,*v*P.

(7) **Move Along Increasing *fseq*-indices (MAIF):** XP can move from A to B iff the heads crossed form a well-formed functional sequence fragment.

Combining unlocking-via-movement and MAIF, I propose that in the absence of successive cyclic movement, Hungarian achieves the transparency of clauses via the *size-dependent adjunction* of embedded clauses to matrix projections (for a similar proposal, see Egressy to appear). In (8), while MAIF allows the embedded TP′ to freely move within the matrix clause, MAIF only allows sub-extraction from TP′ if adjoins to an equal-sized projection, which is here the matrix TP. After the embedded TP′ has adjoined to the matrix TP, XP can move to Spec,CP along the well-formed V′-*v*′-T′-C sequence. That is, upon adjunction to an equal-sized matrix projection, the sequence of heads of the embedded clause finds a continuation in the matrix clause. Size-dependent opacity is derived from the idea that the unlocking-via-adjunction mechanism is size-sensitive: Every clause must adjoin to an equal-sized matrix projection; before adjunction, no movement can leave clauses. Therefore, smaller embedded clauses can become transparent via adjunction earlier (i.e. more matrix positions are accessible from them), whereas larger embedded clauses can only become transparent via adjunction later (i.e. less matrix positions are accessible from them). Position-dependent opacity is derived under the assumption the merger of equal-sized projections causes co-projection and intermediate projections cannot move (e.g. Chomsky 1995). In (8), the embedded TP cannot leave its TP-sister position because it is not a maximal projection anymore. In other words, merger with an equal-sized matrix projection gives transparency at the cost of mobility.

(8)



The discussion proceeds as follows: In Section 2, I present the findings of Egressy (2025) on Williams Cycle effects in Hungarian. Section 3 introduces novel data on the basis of which I claim that Hungarian also exhibits position-dependent opacity. Section 4 demonstrates that position-dependent opacity cannot be reduced to size-dependent opacity, and thus previous accounts of size-dependent opacity do not straightforwardly derive position-dependent opacity. In Section 5, I lay out the details of the unified analysis sketched above.

2 Size-dependent opacity in Hungarian

2.1 The Hungarian left-periphery

This subsection introduces the Hungarian clausal left-periphery summarized in (9): PartP hosts verbal particles; Foc(us)2P and Foc(us)1P host foci; Com(mitment)P hosts sentential adverbs; and Rel(ative)P hosts relative pronouns (Egressy 2025).

The projections in (9) will be discussed in a lower-to-higher order.

(9) $[_{\text{RelP}}$ relative pronoun $[_{\text{ComP}}$ sentential adverb $[_{\text{Foc2P}}$ focus $[_{\text{Foc1P}}$ focus $[_{\text{PartP}}$ particles[VP]]]]]

The lowest relevant left-periphery position is the position of verbal modifiers including particles (e.g. Horvath 1981; Koopman and Szabolcsi 2000; É. Kiss 2006). For instance, the particle *keresztül* 'across' can move to this pre-verbal position in (10). The particle *keresztül* is base-generated as a postposition in a post-verbal and VP-internal PP, where it assigns superessive case to its complement. In Dékány and Hegedűs' (2015) analysis, particle movement is treated as phrasal movement rather than head movement: After *keresztül*'s complement has moved via some Spec,*p*P edge position, the PP-remnant containing only *keresztül* can independently move out of the postpositional extended projection and land higher in the clause. Following Egressy (2025), the landing site of particle movement is referred to as Spec,PartP.

(10) $[_{\text{PartP}}$ ✓ $[_{\text{VP}}[_{\text{PP}}$ … XP …]]]

$[_{\textit{PART}}$ *Keresztül*$]_1$ *fut-ott-am* $[_{\textit{PP}}$ *a* *város-on* $\_\__1]$
across run-PST-1SG.INDF the city-SUP
'I ran across the city.'

Foci and *wh*-constituents both have to move to a position preceding the particle. In the literature, these have been assumed to land in the same position, which has been generally referred to as Spec,FocP (e.g. Horvath 1981, 1986; É. Kiss 1987; Brody 1990). Focus movement is exemplified in (11). Whenever the focused constituent lands in Spec,FocP, the verb moves to the Foc head and therefore precedes the particle (Brody 1990).[2]

(11) $[_{\text{FocP}}$ ✓ Foc+Part+V$[_{\text{PartP}}$ Part+V $[_{\text{VP}}$ V … XP …]]]

$[_{\textit{FOC}}$ *(CSAK) A VÁROS-ON*$]_2$ *fut-ott-am*$_1$ $[_{\textit{PART}}$ *keresztül*$]_3$ $\_\__1$ $\_\__2$ $\_\__3$
only the city-SUP run-PST-1SG.INDF across
'I have ran across ONLY THE CITY '

Hungarian clauses can have multiple foci as in (12) (É. Kiss 1998:p.16). É.Kiss (1998) proposes that such constructions have two focus projections, where Foc1P is dominated by Foc2P, headed by Foc1 and Foc2, respectively. In (12), the verb undergoes head-movement first from Part to Foc1 then from Foc1 to Foc2 in a way that it ends up between the two foci.

(12) $[_{\text{Foc2P}}$ XP [Foc2+✓ $[_{\text{Foc1P}}$ YP [Foc1+T+V $[_{\text{PartP}}$ particle $[_{\text{PartP}}$ ~~T+V~~ … ~~XP~~ … ~~YP~~]]]]]]

[2]Hungarian *wh*-movement lands in the same Spec,FocP position and triggers the same verb-particle inversion (Horvath 1986), and follows the same opacity pattern and focusing (Egressy 2023, to appear).

[$_{FOC2}$ CSAK KÉT LÁNY]$_1$ *olvas-ott-∅*$_2$ [$_{FOC1}$ CSAK EGY KÖNYV-ET]$_3$ __$_2$ [$_{PART}$ *el*] __$_2$ __$_1$ __$_3$
only two girl read-PST-3SG.INDF only one book-ACC away

a vizsgá-ra
the exam-SUB

'ONLY TWO GIRLS were such that they read ONLY ONE BOOK for the exam.'

The distinction between the two focus positions is only possible in clauses with at least two foci. Therefore, in structures with a single focus, I will not make a distinction between landing in Spec,Foc1P and Spec,Foc2P, but use the neutral term Spec,FocP.[3]

The next highest position is the position of sentential adverbs like *szerencsére* 'fortunately', *sajnos* 'unfortunately', and *következésképpen* 'consequently', which always precede foci in the left-periphery as in (13) (Horvath 1981; É. Kiss 1987; Egedi 2021). Krifka (2023) introduces three projections JudgeP, Com(mitment)P, and ActP to host high sentential adverbs. Following Egressy (2025), I assume that all high sentential adverbs are hosted in Spec, ComP. The ordering of these adverbs with respect to each other will not be central to this paper.[4]

(13) [$_{FOC}$ MA] *hall-ott-am,* [*hogy* ($_{SADV}$ *szerencsére)* [$_{FOC}$ (CSAK) A VÁROS-ON]$_2$ *fut-ott-ak*$_1$
today hear-PST-1SG.DEF that fortunately only the city-SUP run-PST-3PL.INDF

[$_{PART}$ *keresztül*]$_3$ __$_1$ __$_2$ __$_3$].
across

Without *szerencsére*: 'I heard that they ran across ONLY THE CITY.' (i.e. the noise of running)

With *szerencsére*: 'I heard that someone said that fortunately they ran across ONLY THE CITY.'

Similarly to languages such as Italian (Rizzi 1997) and Wolof (Torrence 2013), Hungarian has a dedicated projection for relative pronouns, which is not shared by other operators like foci and *wh*-phrases (Horvath 1981). As demonstrated in (14), while foci follow high sentential adverbs, relative pronouns precede them. Furthermore, unlike the movement of foci, the movement of relative pronouns is not accompanied by head movement of the verb. Following Egressy (2025), the position of relative pronouns will be referred to as Spec,RelP. (Throughout the paper, angled brackets refer to alternative positions for a constituent.)

[3] In this paper, clauses with more than two foci are not discussed. Foc1P and Foc2P could host more than two foci if only the highest focus occurs in Spec,Foc2P while all the other foci occur in Spec,Foc1P. At the same time, I do not know of any principled ground to exclude further projections like Foc3P, Foc4P etc.

[4] Further cross-linguistic support comes from Cinque's hiearchy (Cinque 1999), in which evaluative adverbs are hosted in the Mood-Evaluative projection, which is one of the highest projections of the clause.

(14) [$_{\text{RelP}}$ ✓ [$_{\text{ComP}}$[$_{\text{FocP}}$[$_{\text{PartP}}$ [$_{\text{VP}}$[$_{\text{PP}}$ … XP …]]]]]

Ez *az* *a* *város,* ⟨$_{SADV}$ **következésképpen/* **szerencsére*⟩ [$_{REL}$ *amelyik-en*]$_1$ ⟨$_{SADV}$ *következésképpen/*
this that the city, consequently/ fortunately which-SUP consequently/

szerencsére⟩ [$_{FOC}$ *CSAK* *PETI-VEL*] *fut-ott-∅*$_2$ [$_{PART}$ *keresztül*]$_3$ __$_2$ __$_1$ __$_3$
fortunately only Peti-INS run-PST-3SG.INDF across

'This is the city, which consequently/fortunately I ran across ONLY WITH PETI.'

In sum, the clausal functional sequence at this point looks as follows:

(15) Rel>Com>Foc2>Foc1>Part>V

The reason for focusing on the heads in (15) and their projections is that the operators associated with them reliably represent structural height in the left-periphery. For example, sentential adverbs consistently precede foci, so it is reasonable to assume that a dedicated ComP dominates the dedicated FocPs. Before concluding the section, I briefly argue that there are no dedicated projections for topics and quantifiers, that is, (15) lacks the Top(ic)/Ref(erential) and Q(uantifier)/Dist(ributive).

The main motivation for ignoring topics and quantifiers is that only constituents with invariant structural height indicate the structural size of clauses reliably. In Hungarian, topics and positive quantifiers can occur before or after high sentential adverbs in neutral sentences, or even post-verbally (É. Kiss 2002; Egedi 2021):

(16) ⟨[$_{TOP}$ *A* *diák-ok-at*]/ [$_{Q}$ *sok* *diák-ot*]⟩ [$_{SADV}$ *sajnos*] ⟨[$_{TOP}$ *a* *diák-ok-at*]/
the student-PL-ACC many student-ACC unfortunately the student-PL-ACC

[$_{Q}$ *sok* *diákot*]⟩ *el* *kell* *utasít-an-unk* ⟨[$_{TOP}$ *a* *diák-ok-at*]/ [$_{Q}$ *sok* *diákot*]⟩
many student-ACC away need reject-INF-1PL the student-PL-ACC many student-ACC

'{The students/Many students}, we have to reject unfortunately.'

Furthermore, although quantifiers and topics are similar to foci in that they can occur in in finite clauses like (17), there is an essential difference: unlike foci, quantifiers and topics can also occur in radically truncated clauses lacking external arguments, verb inflection, and accusative case marking as (18) (Halm 2021). With Halm (2021), I analyze radically truncated clauses as bare, head-final VPs.

(17) {[$_{TOP}$ *A* *kocsmá-ban*]/ [$_{Q}$ *mindenhol*]/ [$_{FOC}$ *CSAK* *A* *KOCSMÁ-BAN*]} *hangosít-juk* *fel* *a* *TV-t*
the pub-IN everywhere only the pub-IN add.volume-1PL up the TV-ACC.

'{ In the pub/ everywhere/ ONLY IN THE PUB}, we turn up the volume on the TV.'

(18) [{ topics/quantifiers/*foci} [$_{\mathrm{VP}}$ [...]]]

{[$_{TOP}$ *Kocsmá-ban*]/ [$_{Q}$ *mindenhol*]/ [$_{FOC}$ **CSAK KOCSMÁ-BAN*]} *TV fel hangosít*
pub-IN everywhere only pub-IN TV up add.volume
'{ In the pub/ everywhere/ *ONLY IN THE PUB}, (someone) turns up the volume on the TV.'

On the basis of these, I follow Surányi (2002), Szendrői (2003), and É. Kiss (2010) in claiming that while particles, foci, sentential adverbs, and relative pronouns have dedicated projections in the left-periphery, quantifiers and topics do not occur in the specifiers of dedicated projections (i.e. there is no TopP and QP) but can adjoin to/merge with various maximal projections.[5] Throughout the paper, the adjunction site of topics and quantifiers will be assumed to be maximal ComP and FocP projections in finite clauses like (17) (i.e. I set aside cases like (18)). It is a robustly established fact in the Hungarian literature (e.g. É. Kiss 1987, 1998, 2002; Szendrői 2003; É. Kiss 2010) that in the left-periphery no topics and positive quantifiers are c-commanded foci. Here, focus-topic-focus and focus-quantifier-focus orders are excluded by the assumption that the lowest adjunction site for topics and quantifiers is the outermost FocP. Topics and quantifiers will be discussed in greater detail in Section 4. (For a more extended discussion of topics and quantifiers, see Egressy (2025).)

2.2 Clause sizes in Hungarian

Following Egressy (2025, to appear), I assume that the Hungarian clause structure does not have *one dedicated* CP projection for the complementizer. I follow Manetta's (2006; 2011) proposal for the Hindi-Urdu *ki* 'that', and claim that *hogy* 'that' is not a complementizer head in the clausal *fseq* but rather an edge marker, possibly inserted after spell-out. Instead of having a dedicated head in the clausal *fseq*, *hogy* 'that' will be handled as an adjunct to the maximal projection of clauses. For example, if an embedded clause has a sentential adverb but no relative pronoun, it is a bare ComP as in (19), whereas if it has a particle but no other left-periphery operator, it is a bare PartP as in (20).[6] This also means that the sister of the matrix verbs in (19) and (20) is not a CP but a ComP and a PartP, respectively.

(19) [$_{\mathrm{VP}}$ V [$_{ComP}$ *hogy* [sentential adverb [Com [$_{\mathrm{Foc2P}}$ ...]]]]]

(20) [$_{\mathrm{VP}}$ V [$_{\mathrm{PartP}}$ *hogy* [particle [Part [$_{sMod}$ ...]]]]]

There are independent, Hungarian-specific arguments for handling *hogy* 'that' as an edge marker rather than a C(omplementizer) head selecting for a certain category of complement clauses. For example, Kenesei and Szeteli (2022 p. 98) bring a number of naturally occurring examples like (21) in which *hogy* precedes infinitival clauses.

[5] To be more precise, since negative existential quantifiers behave in the same way as foci (É. Kiss 1998), only quantifiers not belonging to this group will be assumed to be constituents lacking dedicated projections. In the case of topics, all subtypes will be assumed to be adjuncts.
[6] An alternative would be to adopt the idea in Radford (2018), according to which *that* occurring at the edge of English clauses of various sizes is always the realization of the highest head of the clause, possibly in a way that *hogy* keeps moving as long as the embedded left-periphery projects.

(21) *Egy élet-et le-het-ne ar-ra áldoz-ni,* [*hogy az-ok-at meg ír-ni*]
a life-ACC be-POT-COND it-SUB sacrifice-INF that that-PL-ACC PART write-INF
'One could sacrifice a life for writing them.'

Furthermore, *hogy* 'that' can introduce either finite or subjunctive clauses, and even embedded questions (Koopman and Szabolcsi 2000). In short, *hogy* does not seem to select for a certain clause-type or a certain syntactic category.

Secondly, *hogy* 'that' often occurs in front of constituents that are clearly non-propositional. Example (22) indicates that reaction words like *hoppá* 'wow' and *brrr* 'brrr' as well as greetings like *helló* and *szia* 'hello'. Although the syntactic category of these words remains an open question, it is clear that they do not involve predication of any kind. Notice that this is different from the behaviour of English *that*, which is not compatible with greetings, e.g. **John did not say that hi.*.

(22) *Győző az-t (se) mond-t-a,* [*hogy brrr/ hoppá/ helló/ szia*]
Győző it-ACC neither say-PST-3SG.DEF that brrr wow hello hello
'Győző did (not even) say hi/ goodbye/ wow /brrr.'

Thirdly, *hogy* 'that' does not occur exclusively in complement clauses: In (23), *hogy* occurs in an adjunct clause preceding the conditional complementizer *ha* 'if'.

(23) *Örül-ök,* [*hogy ha jö-ssz*]
be.happy-1SG.INDF that if come-2SG.INDF
'I am happy if you come.'

Finally, the clause in which *hogy* occurs does not have to be a subordinate clause. Thus, *hogy* 'that' often occurs at the edge of *echo questions* like (24) (Anikó Lipták, p. c.), where it also precedes *wh*-phrases.

(24) *A: Mari jön-∅ este. B: Hogy ki jön este?!*
Mari come-3SG.INDF evening that who come-3SG.INDF evening
A: 'Mari comes tonight.' B: 'Who comes tonight?!'

Thus, the morpheme *hogy* 'that' does not only occur at the edge of a single type of complement clauses, but shows a much wider distribution: The constituent following *hogy* does not have to be tensed, finite, or even propositional, and the constituents containing *hogy* do not have to be the complements of verbs. Non-projecting complementizers are not uncommon in the cross-linguistic literature, either: Embedded clausal spines base generated without a projecting complementizer have been independently proposed by Koopman and Szabolcsi (2000) for Hungarian, by Manetta (2006; 2011) for Hindi-Urdu, Todorović and Wurmbrand (2020) for Bosnian/Croatian/Serbian, Williams (2013) for English, by Angelopoulos (2019) for Greek, and by Kayne (2000) for a number of other languages. In essence, my treatment of *hogy* is somewhat similar to Wurmbrand's (2001) treatment of German *zu*, which does not unambiguously indicate constituent-size, or Kim's (2024) treatment of Korean complementizer *-ko*, which on its own does not add to the opacity of the embedded clause.

In the absence of a projection headed by *hogy* in the left periphery, Egressy (2025, to appear) assumes that clauses with different amounts of material in their left-periphery are of different sizes. Thus, one may say that clauses introduced by *hogy* are not uniformly headed by *hogy*, but by the various heads of the left periphery, which can be referred to as the *exploded CP domain* (Rizzi 1997). Therefore, the largest clauses discussed here are RelPs; these are clauses with a relative pronoun in Spec,RelP. The second largest clauses are ComPs; these are clauses which have sentential adverbs but no relative pronouns. Clauses whose highest left-periphery operators are foci are smaller than ComP or RelP: Clauses with two foci are Foc2Ps; Foc2Ps are larger than clauses with one focus, which are Foc1Ps. Clauses with particles but no relative pronouns, sentential adverbs, or foci are PartPs. This can be summarized as follows:

(25) Clause-sizes in Hungarian: RelP>ComP>Foc2P>Foc1P>PartP>sModP

Egressy (2025) claims that Hungarian subjunctive clauses introduced by *hogy* are structurally smaller than indicative clauses introduced by *hogy*. Furthermore, Egressy (to appear) claims that non-speech-reporting indicative *hogy*-clauses are structurally smaller speech-reporting ones. While these differences between clause type correspond to differences in opacity to various cross-clausal operations, this paper sets these clause-type-differences aside and focuses on the richness of the overt left-periphery, e.g. I will not compare the opacity of indicative and subjunctive clauses. For Egressy (2025), the smallest clauses introduced by *hogy* are subjunctive clauses without particles, foci, sentential adverbs, or relative pronouns. In this paper, I will refer to this clause-type as sModP headed by the subjunctive sMod head. Subjunctive clauses can have the same operators and projections in their left-periphery as indicative clauses (e.g. subjuncitve clauses with sentential adverbs are ComPs).

2.3 Williams Cycle effects in Hungarian

Egressy (2025) argues that the opacity of the various clause-sizes in Hungarian patterns like a version of the *Williams Cycle* (Williams 1974, 2003, 2013) in (26). For now, the Williams Cycle is a descriptive generalization; the theories deriving the Williams Cycle are discussed in Sections 4 and 5.

(26) **Williams Cycle:** Movement to Spec,βP cannot proceed from Spec,αP or across αP, where α>β in the functional sequence. (based on Poole 2022; Williams 2003)

The Williams Cycle has been shown to hold in some form in a number of languages including German (Müller 2014b,a), Swahili (Meadows 2023), Mandarin Chinese (Meadows and Yan 2024), and Georgian (Bondarenko 2024). The best-known illustration of the Williams Cycle is the ban on *hyper-raising* in English (originally observed by Chomsky

1973). For instance, consider (27) in which *John* cannot raise from the embedded CP to the matrix Spec,PartP. Under (26), the ungrammaticality of (27) is expected as C>T in *fseq*.

(27) *John$_1$ seems [$_{CP}$ __$_1$ has eaten an apple]

In contrast, subject raising can proceed from embedded TPs:

(28) John$_1$ seems [$_{TP}$ __$_1$ to have eaten an apple]

The Williams Cycle can be categorized as *size-dependent opacity* as it is the internal structural size of constituents that determines the set of movement types that can proceed from them.

The data on the basis of which Egressy (2025) claims that Hungarian exhibits Williams Cycle effects is summarized in (29). The columns of (29) represent three operation types introduced in Section 2.1: Particle extraction, *wh*-/focus extraction, and relative extraction. The rows represent the clause-sizes introduced in Section 2.2: sModP, PartP, Foc1P, Foc2P, and ComP. In the paragraphs below, I discuss the individual datapoins of (29).

(29) **Opacity depending on clause-size**

	Particle extraction	focus extraction	Relative extraction
sModP	✓	✓	✓
PartP	×	✓	✓
Foc1P	×	✓	✓
Foc2P	×	?	✓
ComP	×	×	✓

The first column in (29) represents particle movement, which was claimed to land in Spec,PartP in Section 2.1. As (30) shows, sModP-sized clauses, which do not have particles, foci, or sentential adverbs are transparent to particle movement (Farkas and Sadock 1989). However, once embedded left-periphery has a particle in (30), i.e. we have evidence that the embedded clause is PartP-sized, particle movement is blocked. Furthermore, as (31) exemplifies, particle movement cannot proceed from clauses with one or two foci, i.e. particle movement is impossible from Foc1Ps and Foc2Ps.[7] Finally, particle movement cannot proceed from ComPs, clauses with adverbs, as shown in (32).

(30) [$_{PartP}$ × … [$_{VP}$ … [$_{PartP}$ particle [… XP …]]]]

[$_{PART}$ *Keresztül*]$_1$ *akar-t-am,* [*hogy* ($_{PART}$ **el)* *fus-s-anak* *a* *város-on* __$_1$]
across want-PST-1SG.INDF that away run-SUBJ-3PL.INDF the city-SUP
'I wanted them to run across the city (*away).'

[7] There are things to note for (31). Firstly, in the first version of (31), the phonologically heavier focus is scrambled after the particle (É. Kiss 2008). Secondly, if no long particle movement happens in the sentence, the matrix clause is more natural with the expletive *azt* in the particle position.

(31) [$_{\text{Part}}$ × … [$_{\text{VP}}$ … ([$_{\text{Foc2P}}$ focus) [$_{\text{Foc1P}}$ focus [… XP …]](])]]

a. ⟨$_{PART}$ **Keresztül*⟩$_1$ *akar-t-am,* [*hogy* [$_{FOC}$ *CSAK* *MARI-VAL*] *fus-s-anak* ⟨$_{PART}$ *keresztül*⟩$_1$ [$_{FOC}$
across want-PST-1SG.INDF that only Mari-INS run-SUBJ-3PL.INDF across
CSAK A VÁROS-ON]$_2$ __$_2$ __$_1$]
only the city-SUP

b. ⟨$_{PART}$ **Keresztül*⟩$_1$ *akar-t-am,* [*hogy* [$_{FOC}$ *CSAK* *MARI-VAL*] *fus-s-anak* ⟨$_{PART}$ *keresztül*⟩$_1$ *a*
across want-PST-1SG.INDF that only Mari-INS run-SUBJ-3PL.INDF across the
város-on __$_1$]
city-SUP

Intended: 'I wanted them to run across {the city/ ONLY THE CITY} ONLY WITH MARI.'

(32) [$_{\text{PartP}}$ × … [$_{\text{VP}}$ … [$_{\text{ComP}}$ sentential adverbs [… XP …]]]]

[$_{PART}$ *Keresztül*]$_1$ *akar-t-am,* [*hogy* ($_{SADV}$ **következésképpen*) *fus-s-anak* *a* *város-on* __$_1$]
across want-PST-1SG.DEF that consequently run-SUBJ-3PL.INDF the city-SUP

'I wanted them to run across the city (*consequently).'

The second column of (29) represents focusing. Unlike particle movement, focus movement can proceed from clauses with particles (PartPs), as in (33). (That sModP-sized clauses are transparent to focusing, too is shown by Example (25b) in (É. Kiss 2002).) Furthermore, focusing can proceed from clauses with a focus, as in (34) (Horvath 1986; É. Kiss 2002). In contrast, (35) shows that clauses with sentential adverbs (i.e. ComPs) are opaque to focus movement in general. As discussed in Footnote 3, the general problem of FocPs is that we only know whether a certain operator moves to Spec,Foc1P, Spec,Foc2P, or even a higher focus projection if there are no further foci in the clause. It is safe to say that particles do block movement to no Spec,FocP, whereas high sentential adverbs block focus movement to all Spec,FocPs. For the sake of simplicity I follow Egressy (2025) in assuming that the matrix landing site in (34) is Spec,Foc2P and the embedded focus is in Spec,Foc1P. That is, focus movement to Spec,Foc2P can proceed from Foc1Ps.

(33) [$_{\text{FocP}}$ ✓ … [$_{\text{VP}}$ … [$_{\text{PartP}}$ particle [… XP …]]]]

[$_{FOC}$ *(CSAK) A VÁROS-ON*]$_2$ *hall-ott-am*$_1$ __$_1$, [*hogy* [$_{PART}$ *keresztül*]$_3$ *fut-ott-ak* __$_2$ __$_3$].
only the city-SUP hear-PST-1SG.DEF that across run-3PL.INDF

'I heard that they ran across ONLY THE CITY.'

(34) [$_{\text{Foc2P}}$ ✓ … [$_{\text{VP}}$ … [$_{\text{Foc1P}}$ focus [… XP …]]]]

[$_{FOC}$ *CSAK A VÁROS-ON*]$_1$ *akar-t-am*$_2$ ___$_2$, [*hogy* [$_{FOC}$ *CSAK MARI-VAL*] *fus-s-anak*$_3$
only the city-SUP want-PST-1SG.DEF that only Mari-INS run-SUBJ-3PL.INDF

[$_{PART}$ *keresztül*]$_4$ ___$_3$ ___$_1$ ___$_4$]
across

'ONLY THE CITY I wanted them to run across ONLY WITH MARI.'

(35) [$_{\text{FocP}}$ × … [$_{\text{VP}}$ … [$_{\text{ComP}}$ sentential adverbs [… XP …]]]]

[$_{FOC}$ *(CSAK) A VÁROS-ON*]$_2$ *hall-ott-am*$_1$ ___$_1$, [*hogy* ($_{SADV}$ *??szerencsére/ ??sajnos)* [$_{PART}$ *keresztül*]$_3$
only the city-SUP hear-PST-1SG.DEF that fortunately unfortunately across

fut-ott-ak ___$_2$ ___$_3$].
run-PST-3PL.INDF

'I heard that (??fortunately/??unfortunately) they ran across ONLY THE CITY.'

The last column in (29) represents relative movement, which lands in Spec,RelP. Cross-clausal relative pronoun movement can proceed from ComP-sized clauses, as in (36), Foc1P/Foc2P-sized clauses, as in (37) (Horvath 1981), or PartP-sized clauses, as in (36). (That clauses without any left-periphery material are transparent to relativization is shown by Example (10a) in (Horvath 1981).) [8]

(36) [$_{\text{RelP}}$ ✓ … [$_{\text{VP}}$ … [$_{\text{ComP}}$ sentential adverbs [… XP …]]]]

Ez az a város, [$_{REL}$ *amelyik-en*]$_1$ *az-t* { *hall-ott-am/ hazud-t-am/ morog-t-am*}, [*hogy*
this that the city which-SUP it-ACC hear-PST-1SG.DEF lie-PST-1SG.DEF growl-PST-1SG.DEF that

($_{SADV}$ *szerencsére)* [$_{PART}$ *keresztül*]$_2$ *fut-ott-ak* ___$_1$ ___$_2$]
fortunately across run-PST-3PL.INDF

'This is the city which I {heard/ lied/ growled} that they (fortunately) ran across.'

(37) [$_{\text{RelP}}$ ✓ … [$_{\text{VP}}$ … [$_{\text{Foc2P}}$ focus [$_{\text{Foc1P}}$ focus [… [$_{\text{PartP}}$ particle [… XP …]]]]]]

Ez az a város, [$_{REL}$ *amelyik-en*]$_1$ *az-t* *akar-t-am,* [*hogy* [$_{FOC}$ *CSAK PETI*] *fus-s-on*$_3$
this that the city which-SUP it-ACC want-PST-1SG.INDF that only Peti run-SUBJ-3SG.INDF

[$_{PART}$ *keresztül*]$_4$ [$_{FOC}$ *CSAK MARI-VAL*] ___$_3$ ___$_1$ ___$_4$]
across only Mari-INS

'This is the city across which I wanted ONLY PETI to run across ONLY WITH MARI'

[8] Although relative movement in (36) is degraded, most speakers find it better than focus movement in (35) on the basis of my informal survey mentioned in Footnote 1.

Despite its freedom compared to other movement types, relative movement cannot proceed from every clause type; it is blocked from relative clauses, i.e. RelPs:

(38) $[_{\text{RelP}} \times \ldots [_{\text{VP}} \ldots [_{\text{RelP}}$ relative pronoun $[\ \ldots$ XP $\ldots\ ]]]]$

**Ez az a város,* $[_{\textit{REL}}$ *amelyik-en*$]_1$ *lát-t-ad az ember-t,* $[$ *hogy* $[_{\textit{PART}}$ *keresztül*$]_2$ *fut-ott-∅* $\_\_{}_1$ $\_\_{}_2]$
this is the city which-SUP see-PST-2SG.DEF the man-ACC that across run-PST-3SG.INDF
Intended: 'This is the city which you saw the man who ran across.'

Even though the literature occasionally posits constraints on Hungarian movement types independently (e.g. for particle movement, see Farkas and Sadock 1989), I adopt the attitude of Starke (2001), Abels (2012), and Egressy (2025) by handling all movement types under a *single descriptive generalization.* This approach includes the idea that opacity and islandhood are not defined in absolute terms but are always relativized to individual movement types, e.g. relative clauses (RelPs) are not handled as absolute islands.

The pattern in (29) is described by Egressy (2025) as the *Strict Williams Cycle* in (39). Similar to the classic Williams Cycle in (26), the Strict Williams Cycle does not allow movement from a certain clause to a lower position. However, the Strict Williams Cycle is more restrictive than the classic Williams Cycle as it does not allow sub-extraction from βP to another Spec,βP. Hence, the smallest of the Hungarian clauses, sModPs, are transparent to the sub-extraction of particles, foci, and relative pronouns because sMod is lower in the *fseq* than any other heads. PartPs (i.e. clauses with particles) are transparent to focusing and relativization but not to particle movement as Foc1, Foc2, and Rel are higher than Part in *fseq* but Part is not. Foc1Ps and Foc2Ps, clauses with one and two foci, are uniformly transparent to relativization because Rel>Foc2>Foc1 in *fseq*, but uniformly opaque to particle movement because Foc2>Foc1>Part in *fseq*. Focusing from Foc1Ps is possible because it can land in a higher Spec,Foc2P. (Since the complications related to the upper limit on the number of focus projections is set aside, focusing from Foc2Ps is not discussed.) Clauses with sentential adverbs are ComPs, which are transparent to relative movement because Rel>Com in *fseq* but opaque to focusing and particle movement because Com>Foc2>Foc1>Part in *fseq*. Finally, relative clauses, RelPs, are islands to every movement type because there is no movement landing in a higher projection than RelP. That the same pattern extends to examples involving multiple embedding is shown in (Egressy 2023:p.20).

(39) **Strict Williams Cycle/Size-dependent opacity:** Movement to Spec,βP cannot proceed across a dominating αP, where $\alpha \geq \beta$ in the functional sequence.

Whether all instances of the classic Williams Cycle in (26) can be re-analyzed with the strict Williams Cycle in (39), or the two patterns co-exist in the languages of the world is set aside. It needs to be highlighted that Hungarian is not the only attested instance of the Strict Williams Cycle pattern, i.e. a pattern where the ban on movement extends to the same position in the next lowest clause. One example comes from German. In German, cross-clausal scrambling, which lands in Spec,TP (Keine 2016, 2019, 2020), cannot proceed from verb-final CPs introduced by *dass* 'that', as in (40) (Müller

2014a,b). In contrast, cross-clausal topicalization, which lands in Spec,ForceP can proceed from CPs (Keine 2020:p.219), as in (41). As far as (40) and (41) are concerned, the German pattern can be described with either the classic or the strict Williams Cycle if the German *fseq* is Force>C>T. However, it is covered by the strict but not the classic Williams Cycle that relativization, which lands in Spec,CP, cannot proceed from CPs, as shown in (42) (Keine 2020:p.219).[9] Further instances of the Strict Williams Cycle pattern are exhibited by the Italian FocP, from which no movement to Spec,FocP can proceed, whereas relative movement to Spec,ForceP is possible (Force>Foc in *fseq*) (Rizzi 1997; Abels 2012); by the Hindi CP, which is opaque to *wh*-licensing driven from higher C heads (Keine 2016, 2020); and by German TP-sized infinitival clauses, which are opaque to scrambling to Spec,TP (Keine 2016, 2020).

(40) $[_{\text{TP}}$ × … $[_{\text{VP}}$ … $[_{\text{CP}}$ [… XP …]]]]

dass$[_{TP}$ ⟨**das Buch*⟩$_1$ *Karl* _$_2$ *glaub-t* $[_{CP}$ *dass* $[_{TP}$ ⟨*das Buch*⟩$_1$ *keiner* _$_1$ *lies-t*$]_2]]$
that the book Karl.ACC think-3SG that the book.ACC no.one.NOM read-3SG
'that Karl thinks that Peter reads the book.'

(41) $[_{\text{ForceP}}$ ✓ … $[_{\text{VP}}$ … $[_{\text{CP}}$ … XP …]]]

[*Die Maria*$]_2$ *hab-e* *ich* __$_1$ *gedacht* $[_{CP}$ *dass der Fritz* __$_2$ *lieb-t*$]_2$.
the Maria.ACC have-1SG I think.PTCP that the Fritz.NOM love-3SG
'Maria, I thought that Fritz loves.'

(42) $[_{\text{CP}}$ × … $[_{\text{VP}}$ … $[_{\text{CP}}$ … relative pronoun …]]]

**der Junge* $[_{CP}$ *den*$_1$ *sie glaub-t* $[_{CP}$ *dass der Peter* __$_1$ *getroffen ha-t*]]
the boy who.ACC she believe-3SG that the Peter meet.PTCP have-3SG
Intended: 'The boy that she believes that Peter has met.'

My proposal in Section 5 will be concerned with the derivation of the strict Williams Cycle in (39) and will argue that (39) arises in Hungarian because of the absence of successive cyclic movement in the clausal domain.

3 Position-dependent opacity in Hungarian

3.1 The right- and leftward movement of clauses

Following the mainstream literature, I assume that clauses introduced by *hogy* are base generated within the matrix VP, where they are locally selected as the complement of the matrix V. However, the adjuncts *tegnap* and *egy buliban* obligatory occurring between the verb and post-verbal embedded clause suggest that the embedded clause must undergo

[9] Furthermore, Keine (2020) mentions that for a group of speakers *wh*-movement to the Spec,CP of embedded questions cannot proceed from CPs, either.

obligatory extraposition to the right (É. Kiss 2002). Based on examples like (43) and further examples involving noun-complement clauses, É. Kiss's (2002) draws the generalization that *hogy*-clauses cannot be internal to a lexical projection (É. Kiss 2002:p. 237).

(43) [[[$_{\text{VP}}$ V [*hogy* …]] adjuncts] ✓]

[$_{\textit{FOC}}$ *Csak* *ő*$_{i/*j}$] *mond-t-a* __$_1$ ⟨ *tegnap*⟩ ⟨ *egy* *buli-ban*⟩, [*hogy* *Mari*$_j$ *alsz-ik*]$_1$ ⟨ * *tegnap*⟩ ⟨ * *egy* *buliban*⟩.
only she refer-PST-3SG.INDF yesterday a party-IN that Mari sleep-3SG.INDF yesterday a party-IN

'Fortunately, only she$_{i/*j}$ referred (to the fact) yesterday in a party that Mari$_j$ was sleeping'

The syntactic and semantic reasons behind clausal extraposition have been discussed extensively in the cross-linguistic literature (e.g. Moulton 2015; Bruening 2018). Here I follow the Final-over-Final-based proposal by Bondarenko (2023) on Azeri. As Halm (2021) argues, Hungarian VPs are head-final because this is the order that we find in VP-sized radically truncated clauses like (18). In contrast any finite clauses, which are here sModPs, PartPs, Foc1Ps, Foc2Ps, ComPs, and RelPs, are head-initial. Assuming these clausal projections and the VP belong to the same extended projection, if either of these head-initial clausal projections occur within the head-final VP-projection, the Final-Over-Final Constraint in (44) is violated (Biberauer et al. 2014; Sheehan et al. 2017). Bondarenko (2023) assumes that (44) needs to be satisfied at PF, and only the pronounced copies of clauses are considered. That is, unless the clause is removed from the VP in another way, finite clauses/*hogy*-clauses must leave their VP-internal position and right-adjoin to a higher, head-initial projection before the matrix clause gets spelled out. As I will further discuss in Section 5.4, the adjunction of clauses can happen to sModP or any higher clausal projection.

(44) **Final-over-Final Constraint:** A head-final phrase αP cannot dominate a head-initial βP, where α and β are heads in the same extended projection.

One case in which an independent movement type avoids the violation of (44) is the topicalization of clauses. In (45), the embedded clause adjoins in one of the topic-positions introduced in Section 2.1. If it is below sentential adverbs, its position is a FocP-adjunct position, if it is above sentential adverbs, its position is a ComP-adjunct position. That topicalization involves movement rather than base-generation is shown by the fact that the clause reconstructs for Principle C to a position c-commanded by the focus in Spec,FocP (Kenesei 1994; Egressy 2023).

(45) [✓ [$_{\text{ComP}}$ sentential adverbs[✓ [$_{\text{FocP}}$ … [$_{\text{VP}}$ V [*hogy* …]]]]]]

⟨$_{TOP}$ *hogy* *Mari$_j$* *ismer-t-e* *Péter-t*⟩$_1$ [$_{SADV}$ *szerencsésre*] ⟨$_{TOP}$ *hogy* *Mari$_j$* *ismer-t-e*
that Mari know-PST-3SG.DEF Péter-ACC fortunately that Mari know-PST-1SG.DEF

Péter-t⟩$_1$ [$_{FOC}$ *CSAK* *ő$_{i/??j}$*] *mond-t-a* [$_{\text{DP}}$ *pro* ___$_1$]
Péter-ACC only she say-PST-3SG.DEF pro.ACC

'As for that Mari$_j$ had known Péter, fortunately, she$_{i/??j}$ did not tell it.'

Importantly, Section 3.3 will show that Hungarian embedded clauses cannot occur in lower left-periphery positions.

3.2 Position-dependent opacity

Although moved and adjoined clauses have been handled as absolute islands in the earlier syntactic literature (Ross 1967; Wexler and Culicover 1983), the more recent literature has found that subextraction can proceed from moved/adjoined clauses (e.g. Sauerland 1999; Collins 2005; Privoznov 2021). Thus, building on the idea that moved constituents are *not* islands, this section describes how the landing site of clauses in the left-periphery influences sub-extraction possibilities. The generalization drawn in the section will that the movement of the clause cannot be remnant movement.

As Section 3.1 described, clauses can occur in topic positions, which I defined as sisters to maximal FocPs or ComPs. Example (46) shows that even if a clause is transparent to the subextraction of particles in a post-verbal position, particle extraction is not compatible with the subsequent topicalization of the clause. That is, even if we do not expect the impossibility of particle movement on the basis of *clause-internal size* based on Section 2, the clause's position can introduce blocking effects. Notice that it does not matter whether the clause is directly pre-verbal, i.e. it is sister to a maximal FocP, or it precedes a sentential adverb, i.e. it is sister to a maximal ComP.

(46) [$_{\text{ComP/FocP}}$ … [*hogy* … ~~XP~~] … [$_{\text{PartP}}$ × [… [$_{\text{VP}}$ V ~~[*hogy* … XP]~~]]]]

⟨$_{TOP}$* *hogy* ___$_2$ *fus-s-ak* *a* *város-on*⟩$_1$ ($_{SADV}$ *szerencsére)* [$_{PART}$ *keresztül*]$_2$ *akar-t-a* ___$_1$,
that run-SUBJ-1SG.INDF the city-SUP fortunately across want-PST-3SG.DEF

⟨ *hogy* ___$_1$ *fus-s-ak* *a* *város-on*⟩$_1$
that run-SUBJ-1SG.INDF the city-SUP

'(Fortunately,) she wanted me to run across the city.'

The example in (47) shows that focus subextraction is also incompatible with the subsequent topicalization of the clause. Again, it does not matter whether the clause precedes an adverb (i.e. whether it adjoins to FocP or ComP).

(47) [$_{\text{ComP/FocP}}$ … [*hogy* … ~~XP~~] … [$_{\text{FocP}}$ × [… [$_{\text{VP}}$ V ~~[*hogy* … XP]~~]]]]

⟨$_{\textit{TOP}}$* *hogy* [$_{\textit{PART}}$ *keresztül*]$_3$ *fut-ott-am* __$_2$ __$_3$⟩$_1$ ($_{\textit{SADV}}$ *szerencsére)* [$_{\textit{FOC}}$ *CSAK A VÁROS-ON*]$_2$
that across run-PST-1SG.INDF fortunately only the city-SUP

hall-ott-a __$_1$, ⟨ *hogy* [$_{\textit{PART}}$ *keresztül*]$_3$ *fut-ott-am* __$_2$ __$_3$⟩$_1$
hear-PST-3SG.DEF that across run-PST-1SG.INDF

'She heard that I ran across ONLY THE CITY.'

However, topic clauses adjoined to matrix maximal FocP or ComP projections are not islands; relative pronouns can be subextracted from them, as in (48). The embedded clause can precede or follow sentential adverbs i.e. it can adoin to FocP or ComP. Notice that the matrix-modifying sentential adverb showing up between the subextracted relative pronoun and the embedded clause of origin rules out clausal pied-piping as an option because it shows that the relative pronoun leaves the embedded clause.

(48) [$_{\text{RelP}}$ ✓ … [$_{\text{ComP/FocP}}$ … [*hogy* … XP] … [$_{\text{FocP}}$ … [$_{\text{VP}}$ V ~~[*hogy* … XP]~~]]]]]

Ez az a város, [$_{\textit{REL}}$ *amelyik-en*]$_2$ ⟨$_{\textit{SADV}}$ *sajnos*⟩ [$_{\textit{TOP}}$ *hogy* [$_{\textit{PART}}$ *keresztül*]$_3$ *fut-ott-am* __$_2$
this that the city which-SUP unfortunately that across run-PST-1SG.INDF

__$_3$]$_1$ ⟨$_{\textit{SADV}}$ *sajnos*⟩ [$_{\textit{FOC}}$ *CSAK MARI*] *hall-ot-a* __$_1$.
unfortuantely only Mari hear-PST-3SG.DEF

'This is the city which, that I ran across, only Mari heard it unfortunately.'

(Context: I wanted it to be heard by more people)

In sum, clauses that later adjoin to ComP or FocP are opaque to particle and focus subextraction to matrix Spec,PartP and Spec,FocP, respectively but topicalization can be followed by relative extraction to Spec,RelP. The generalization in (49) summarizes these facts: A clausal projection αP cannot undergo (remnant) movement to a left-periphery position immediately dominated by βP if movement originating in αP lands in a lower left-periphery position Spec,γP (i.e. if $\beta \geq \gamma$). For example, since FocP/ComP-sister topic clauses c-command Spec,FocP and Spec,PartP, if focus or particle moves out from some clause, the clause cannot be topicalized. (Since (62) excludes any leftward-moved clause c-commanded by any focus, the Foc1P-Foc2P-distinction is irrelevant; every topic clause c-commands each foci.) In contrast, Spec,RelP c-commands the adjunction sites of topic clauses, hence, (49) does not constrain relative movement from topic clauses.[10]

[10] Anikó Lipták (p.c.) asks whether position-dependent opacity must be defined in terms of hierarchy, or whether the linear definition "clause of origin cannot precede subextracted material" would suffice. One possible argument against this linear formulation and in favour of my structural formulation comes from the movement of N(egative) C(oncord) I(tems). According to É. Kiss (2010), NCIs must adjoin to the NegP/PolP headed by the negative Neg. Embedded NCIs cross the clause boundary and adjoin to either to the right or left of the matrix NegP/PolP as in (i). If the clause is post-verbal, the rightward NCI-movement lands behind (and hierarchically above) the embedded clause, which is incompatible with the linear but compatible with the hierarchical formulation of position-dependent opacity.

(49) **Position-dependent opacity**: αP cannot move to a position immediately dominated by a βP, if some XP from αP moved to Spec,γP, and α, β, γ ∈ *fseq* F, and β ≥ γ.

$$[_{\beta P} \times \ldots [_{\gamma P}\ \mathrm{XP}\ [\ \ldots [_{VP}\ \mathrm{V}\ \boxed{[_{\alpha P}\ \textit{hogy} \ldots \cancel{\mathrm{XP}}\ ]}]]]]$$

Another example for (49) comes from German. At least for certain German speakers, *wh*-movement can proceed from embedded CPs to Spec,CP, as in (50) (Müller 2014a,b; Keine 2016, 2019, 2020). However, after *wh*-subextraction has taken place from it, the embedded CP cannot undergo subsequent topicalization to a higher Spec,ForceP, as in (51) (Müller 1999:p.25). This is striking because embedded CPs can be topicalized otherwise (Müller 1999). This is parallel to (49) if Force and C are both members of the clausal *fseq* in German and Force>C in *fseq*.

(50) *Ich weiss nicht* [$_{CP}$ *wen*$_1$ *er* __$_2$ *gesagt hat* [$_{CP}$ *dass Fritz* __$_1$ *liebt*]$_2$]
I knew.1SG not who.ACC he said have.3SG that Fritz love.3SG
'I did not know who he said that Fritz loves.'

(51) *[$_{CP}$ *Dass Fritz* __$_1$ *liebt*]$_2$ *weiss ich nicht*[$_{CP}$ *wen*$_1$ *er* __$_2$ *gesagt hat*]
that Fritz love.3SG know.1SG I not who.ACC he said have.3SG
Intended: 'I do not know who he said that Fritz loves.'

The generalization in (49) is an extended-projection-specific a ban on remnant movement. That is, it is a restricted version of the Generalized Proper Binding Condition (Fiengo 1977; Lasnik and Saito 1994), which claims that traces must be bound throughout a derivation. Importantly, (49) does not claim that all cases of remnant movement are ungrammatical. Another comparable generalization to (49) is the *Müller-Takano Generalization* (Müller 1993; Takano 1994), which says that the subextraction from some αP cannot be the same type as the remnant movement of αP. The movement of relative pronouns, topics, quantifiers, foci, and particles can be all categorized as $\overline{\text{A}}$-movement (or movement landing in $\overline{\text{A}}$-positions) in Hungarian. Müller-Takano-style reformulation of (49) would be the following: A clausal constituent with a left-periphery (i.e. topped some $\overline{\text{A}}$-projection) cannot undergo $\overline{\text{A}}$-remnant movement if $\overline{\text{A}}$-subextraction has taken place from within it. In other words, the only way (49) is more restrictive than the Müller-Takano Generalization is that (49) also constrains the type of the remnant itself, not just the type of the movement operations. The closest domain-specific restriction to (49) is claimed for the nominal domain by Cinque (2005): The

(i) *Mari* ⟨$_{POL}$ *'senki-t*⟩$_1$ [$_{PolP}$ *nem akar-t-a*$_2$ __$_2$, [*hogy* [$_{PART}$ *meg*] *puszil-j-∅* __$_1$]] ⟨$_{POL}$ *'senki-t*⟩$_1$.
Mari no.one-ACC not want-PST-3SG.DEF that part kiss-SUBJ-2SG.INDF no.one-ACC
'Mari did not want you to kiss anyone' lit. 'Mari did not want you to kiss no one.'

The movement of NCIs is size-sensitive: Tensed clauses like (ii) are opaque to NCI-extraction. This fits the picture presented in Section 2 if *fseq*=...Part>Pol>sMod.

(ii) **Mari* ⟨$_{POL}$ *'senki-t*⟩$_1$ [$_{PolP}$ *nem* {*hall/hazud*}*-t-a*$_2$ __$_2$, [*hogy* [$_{PART}$*meg*] *puszil-t-tál* __$_1$]] ⟨$_{POL}$ *'senki-t*⟩$_1$.
Mari no.one-ACC not hear/lie-PST-3SG.DEF that PART kiss-PST-2SG.INDF no.one-ACC
Intended: 'Mari did not {hear/ lie} that you kissed anyone.'

projections of the nominal extended projection cannot remnant move to nominal projections after sub-extraction to another nominal projection has taken place from them.

3.3 Embedding with expletives

Similarly to a number of languages, Hungarian finite clauses introduced by *hogy* sometimes co-occur with overt pronominals (e.g. É. Kiss 1987; Kenesei 1994; Horvath 1997; É. Kiss 2023). English occasionally exhibits this behaviour as well e.g. *I doubt it that you come* (Kiparsky and Kiparsky 1971). The Hungarian pronominal *az* 'it' associated with clauses bears the case suffix assigned by the matrix verb. I will refer to *az* 'it' with the term (*clausal*) *expletive* following Kenesei (1994), Horváth (1997), É. Kiss (2002), and Den Dikken (2018). As we will see in this subsection and the next one, embedding with expletives extends on the position-based pattern of Section 3.2.

In (52), the expletive *az* 'it' receives accusative case from the matrix verb within the matrix VP. That expletives can bear various other cases/postpositions depending on the role of the embedded clause in the matrix clause is discussed by Egressy (to appear). That the clausal expletive and the clause form a constituent is shown by the fact that they can move together to a topic position. This topic position can be either a FocP-adjoined topic position below sentential adverbs or a ComP-adjoined topic position above sentential adverbs. That the clause and the expletive do not move separately to these topic positions is shown by the fact that the two cannot be separated by sentential adverbs in (52). Following É. Kiss (2002), I assume that the constituent containing the expletive and the clause is a DP, whose D head is the expletive (for an alternative, see Lipták 1998). (That this not lead to the absolute islandhood of clauses in DPs is shown in the next subsection.) Similar to the topicalization of expletive-less clauses like (45), the topicalization of clauses occurring in expletive-headed DPs involves genuine movement from VP to the left-periphery, as it is shown by reconstruction for Principle C in (52).

(52) [✓ [$_{\text{ComP}}$ sentential adverbs[✓ [$_{\text{FocP}}$ … [$_{\text{VP}}$ V [$_{\text{DP}}$ *az* [*hogy* …]]]]]]]

⟨$_{\textsc{sadv}}$ *Szerencésre*⟩ [$_{\textsc{top,dp}}$ *az-t* ⟨$_{\textsc{sadv}}$?? *szerencésre*⟩ [*hogy* *Mari*$_j$ *ismer-t-e* *Péter-t*]]$_1$ ⟨$_{\textsc{sadv}}$
fortunately it-ACC fortunately that Mari know-PST-3SG.DEF Péter-ACC

szerencésre⟩ [$_{\textsc{foc}}$ *CSAK* *ő*$_{i/??j}$] *mond-t-a* __$_1$
fortunately only she say-PST-3SG.DEF

'As for that Mari$_j$ had known Péter, fortunately, she$_{i/??j}$ did not tell it.'

From expletive-headed DPs in topic positions, clauses can move out to the right periphery, as in (53). The clause only leaves the expletive-headed DP after the DP has reached its final position in the left-periphery. This is supported by (52): the expletive cannot be above the sentential adverb if the clause is below the sentential adverb, that is, the FocP-adjunction of the clause cannot be followed by the ComP-adjunction of the DP-remnant. In short, the clause always c-commands its expletive; either by remaining within the DP, as in (52), or by moving out of the DP after the DP has reached its final position in the left-periphery, as in (53).

(53) [[DP *az* [*hogy* …]] [ComP sentential adverbs[[DP *az* [*hogy* …]] [FocP …[VP V ~~DP~~] … ✓

⟨TOP,DP *Az-t* __2 ⟩1 [SADV *szerencésre*] ⟨TOP,DP *az-t* __2 ⟩1 [FOC *CSAK* *Ő*$_{i/??j}$] *mond-t-a* __1 *tegnap*
it-ACC fortunately it-ACC only she say-PST-3SG.DEF yesterday

egy *buli-ban* [*hogy* *Mari*$_j$ *ismer-t-e* *Péter-t*]2
a party-IN that Mari know-PST-3SG.DEF Péter-ACC

'As for that Mari$_j$ had known Péter, fortunately, ONLY SHE$_{i/??j}$ did not tell it.'

Unlike topic positions, the other left-periphery positions cannot contain embedded clauses: after the expletive-headed DP moves to them, the clause must be immediately extraposed to the right. Firstly, É. Kiss (1981) observed that entire clauses cannot be foci, the movement of clauses cannot land in Spec,FocP. Thus, in (54), neither the embedded clause nor the DP containing the clause and the expletive can occur in Spec,FocP. Furthermore, the fact that the presence of another focus does not make the sentence better shows that embedded clauses cannot occur either in Spec,Foc1P or Spec,Foc2P.

(54) [Foc2P/Foc1P × […[VP V [*hogy* …]]]]]]

*(FOC *CSAK* *EMMÁ-NAK*) [FOC *CSAK* (*AZ-T*) *HOGY* *ISMER-T-EM* *PÉTER-T*]1 *mond-t-am* (FOC *CSAK*
only Emma-DAT only it-ACC that know-PST-1SG.DEF Péter-ACC say-PST-1SG.DEF only

EMMÁ-NAK) __1.
Emma-DAT

Intended: 'I only told (TO EMMA) THAT I HAD KNOWN PÉTER.'

Still, the embedded clause can get a focus-interpretation if the clause leaves the DP in Spec,FocP in a way that only the overt expletive is left behind, as it happened in (53). In the a. version of (55), the DP occurs in Spec,Foc2P; in the b. version, the DP occurs in Spec,Foc1P. Either of these are grammatical if the embedded clause leaves the DP and adjoins in the right-periphery.

(55) $[_{\text{Foc1P/Foc2P}}$ $[_{\text{DP}}$ *az* [*hogy* …]] [… $[_{\text{VP}}$ V ~~DP~~] …]✓

a. $[_{FOC}$ *CSAK AZ-T* $\_\_\__2]_1$ *mond-t-am* $[_{FOC}$ *CSAK EMMÁ-NAK*]
only it-ACC say-PST-1SG.DEF only Emma-DAT

$\_\_\__3$ $\_\_\__1$ [*hogy ismer-t-em Péter-t*$]_2$.
that know-PST-1SG.DEF Péter-ACC

'The only thing such that I told it only to Emma was that I know Peter.'

b. $[_{FOC}$ *CSAK EMMÁ-NAK*] *mond-t-am*$_3$ $[_{FOC}$ *CSAK AZ-T* $\_\_\__2$ $]_1$
only Emma-DAT say-PST-1SG.DEF only it-ACC

$\_\_\__3$ $\_\_\__1$ [*hogy ismer-t-em Péter-t*$]_2$.
that know-PST-1SG.DEF Péter-ACC

'Emma was the only person such that the only thing I told them was that I know Peter.'

Embedded clauses cannot occur in Spec,PartP, either. A subset of Hungarian verbs including *akar* 'want' counts as *stress-avoiding*: These cannot bear the phonetically neutral sentential stress, that is, they cannot be sentence-initial unless they are contrastively or emphatically stressed (Komlósy 1992; Szendrői 2003). Particles can move to Spec,PartP to bear neutral sentential stress, however, entire clauses or DPs containing the clause cannot move to the Spec,PartP particle position, as in (56).

(56) $[_{\text{PartP}}$ × [… $[_{\text{VP}}$ V [*hogy* …]]]]]]

*$[_{PART}$ *(Az-t,) hogy* $[_{PART}$ *keresztül*$]_2$ *fu-s a város-on* $\_\_\__2]_1$ *akar-t-am* $\_\_\__1$
it-ACC that across run-SUBJ.2SG.INDF the city-SUP want-PST-1SG.INDF
Intended: 'I wanted you to run across the city.'

Nevertheless, when the embedded clause strands the expletive-headed DP in Spec,PartP and adjoins in the right-periphery, as in (57), the sentence is grammatical. The remnant of the DP in Spec,PartP works like any other particle, e.g. it becomes post-verbal if the verb undergoes movement to the Foc head in (58). Notice that in (58), the particle must be separated from its expletive. That is, given that clauses must c-command their expletives throughout the derivation, once the the expletive reaches its position in Spec,PartP, the clause must move out to the right from the DP.

(57) $[_{\text{PartP}}$ $[_{\text{DP}}$ *az* [*hogy* …]] [… $[_{\text{VP}}$ V ~~DP~~] …]✓

Peti $[_{PART}$ *az-t* $\_\_\__3]_1$ *akar-t-a* $\_\_\__1$, [*hogy* $[_{PART}$ *keresztül*$]_2$ *fu-s a város-on*$\_\_\__2]_3$
Peti it-ACC want-PST-3SG.INDF that across run-SUBJ.2SG.INDF the city-SUP
'Peti wanted you to run across the city.'

(58) [$_{\text{FocP}}$ focus Foc+✓ [$_{\text{PartP}}$ [$_{\text{DP}}$ *az* [*hogy* …]] T+V […[$_{\text{VP}}$ V ~~DP~~] … ✓

[$_{\textit{FOC}}$ *Csak Peti*] *akar-t-a*$_{4}$ [$_{\textit{PART}}$ *az-t* __$_{3}$]$_{1}$ __$_{4}$ __$_{1}$ *tegnap egy buli-ban,* [*hogy* [$_{\textit{PART}}$
only Peti want-pst-3sg.indf it-acc yesterday a party-in that

keresztül]$_{2}$ *fu-s a város-on* __$_{2}$]$_{3}$
across run-subj.2sg.indf the city-sup

'Yesterday at the party, only Peti wanted you to run across the city.'

Examples like (59) and (60) suggest that if a clause gets to Spec,PartP, Spec,Foc1P, or Spec,Foc2P, it must be moved to the right periphery in the next derivational step. The matrix clauses in these examples contain two expletive-headed DPs: one in Spec,PartP and another in Spec,FocP. While the two DPs are the elative and sublative arguments of *következtet* 'infer' in both examples, the main clause in (59) has the elative expletive-head DP in Spec,FocP and the sublative expletive-headed DP in Spec,PartP; in (59), the roles are flipped: the sublative DP is in Spec,FocP, and the elative DP is in Spec,PartP. Similar examples are provided by É. Kiss (2002:p.238).

(59) [$_{\textit{FOC}}$ *Csak ab-ból* __$_{2}$] *következtet-t-em*$_{1}$ [$_{\textit{PART}}$ *ar-ra* __$_{3}$] __$_{1}$ *tegnap* [*hogy Mari haza men-t-∅*]$_{3}$
only it-el infer-pst-1sg.def it-sub yesterday that Mari home go-pst-3sg.indf

[*hogy nem vol-t-∅ elég lány a buli-ban*]$_{2}$.
that not be-pst-3sg.indf enough girl the party-in

'Yesterday I inferred that Mari left only from the fact that there were not enough girls at the party.'

(60) [$_{\textit{FOC}}$ *Csak ar-ra* __$_{2}$] *következtet-t-em*$_{1}$ [$_{\textit{PART}}$ *ab-ból* __$_{3}$] __$_{1}$ *tegnap* [*hogy Mari haza men-t-∅*]$_{3}$
only it-el infer-pst-1sg.def it-sub yesterday that Mari home go-pst-3sg.indf

[*hogy nem vol-t-∅ elég lány a buli-ban*]$_{2}$.
that not be-pst-3sg.indf enough girl the party-in

'The only thing I inferred yesterday from the fact that Mari went home was that there were not enough girls at the party-'

Essentially, the clauses moving out from these DPs to the right periphery show the mirror image of the order of the DPs in the left-periphery as depicted in (61): the clause moved out from Spec,FocP follows the clause moved out from Spec,PartP. I claim that this shows that once the first DP moves to Spec,PartP, the embedded clause adjoins to the next projection, which is PartP; once the second DP moves to Spec,FocP, the embedded clause adjoins with the next projection, which is FocP. (Again, that the clauses are external to VP is shown by their position after *tegnap* 'yesterday'.)

(61) [$_{\text{FocP}}$ [$_{\text{DP}}$ az [*hogy* …]] [$_{\text{PartP}}$ [$_{\text{DP}}$ az [*hogy* …]] …[$_{\text{VP}}$ …] … ✓ $_{\text{PartP}}$] ✓ $_{\text{FocP}}$]

The constraint in (62) summarizes these facts by claiming that embedded clauses must immediately leave the left-periphery specifiers Spec,PartP, Spec,Foc1P, and Spec,Foc2P. Topic positions are not dedicated specifiers, so (62) does not initiate the removal of clauses from topic positions. É.Kiss (2002:p.88-89) mentions that (62) also holds for DP-internal relative clauses, e.g. relative clauses move out of focused DPs. For the reasons behind (62), I follow Kenesei (1994), who claims that the ban on left-periphery clauses within FocP has a phonological explanation, which is compatible with the assumption that the trace of embedded clauses can still in these positions. Why clauses need to leave the DP in the next derivational step is derived in (62).

(62) If a DP containing an embedded clause introduced by *hogy* occurs in a Spec,αP position in the clausal left-periphery, the clause must adjoin to αP.

For the sake of simplicity, I make two assumptions on the distribution of expletives. Firstly, I follow den Dikken (2018), who assumes that expletives only occur in those structures where they are overtly present; if the clause occurs without an overt expletive, the finite clause is directly taken as a complement by the matrix verb. [11] Secondly, I assume the expletive-based strategy of embedding always involves the expletive-headed DP's movement to a left-periphery position, i.e. no expletives stay VP-internal. After leaving the matrix VP, the DP containing the expletive and the clause can move to a FocP/ComP-adjoined topic position, a Spec,Foc1P/Foc2P focus position, or a Spec,PartP particle position. The structure in (52) has already shown that the expletive-headed DP can move to ComP/FocP-adjoined topic position.

3.4 Expletive-mediated position-dependent opacity

While the positions in which the embedded clauses themselves can be pronounced are highly restricted, the expletive-headed DPs can show up in Spec,PartP, Spec,Foc1P, Spec,Foc2P, and in FocP/ComP-adjoined positions. Since the expletive-headed DPs cannot undergo remnant movement, the position of expletives in the left-periphery allows us to declare of lowest position of clauses in the right-periphery. Building on this idea, this section extends the position-dependent generalization of Section 3.2 to examples involving expletives.

The example in (63) indicates that the subextraction of particle *keresztül* 'across' is only possible if there is no clausal expletive in the structure, i.e. the matrix VP contains no expletive-headed DP. In particular, particle subextraction is incompatible with an expletive-headed DP-remnant in Spec,PartP, Spec,FocP, and in ComP/FocP-adjoined topic positions, as it is indicated by the banned expletive positions in (63). Thus, at least for particle movement, it holds that subextraction is not compatible with overt expletives (cf. Lipták 1998; É. Kiss 2002). (In the bracket diagrams to come, rightward pointing dashed arrows to another left-periphery position indicate that a clause whose expletive shows up in a higher position is earlier opaque to subextraction to a lower position.)

[11] For analyses assuming the general presence of clausal expletives, see Kenesei (1994) and Egressy (2023).

(63) [$_{\text{ComP}}$ [$_{\text{DP}}$ *az* ~~[*hogy* … XP]~~] [$_{\text{FocP}}$ [$_{\text{DP}}$ *az* ~~[*hogy* … XP]~~] [$_{\text{PartP}}$ [$_{\text{DP}}$ *az* ~~[*hogy* … XP]~~] × [… [$_{\text{VP}}$ V …

Mari	⟨$_{\textit{TOP}}$	**az-t*	__$_2$⟩	($_{\textit{SADV}}$	*szerencsére)*	⟨$_{\textit{FOC}}$	**CSAK*	*AZ-T*	__$_2$⟩	⟨$_{\textit{PART}}$ **az-t*	__$_2$⟩	[$_{\textit{PART}}$ *keresztül*]$_1$
Mari		it-ACC			fortunately		only	it-ACC		it-ACC		across

akar-t-a	__$_2$,	[*hogy*	*fus-s*	*a*	*város-on*	__$_1$	]$_2$
want-PST-3SG.DEF		that	run-2SG.SUBJ.INDF	the	city-SUP		

Without expletives: 'Mari wanted you to run across the city.'

Focus movement reveals that overt expletives do not bring about absolute islandhood. As (64) shows, overt expletives in the Spec,PartP particle position are compatible with long focus-extraction from the clause associated with the expletives. At the same time, focus extraction cannot proceed from clauses whose expletive-headed DP subsequently undergoes topicalization, i.e. adjoins to some maximal FocP or ComP position as indicated by the expletive positions in (64).

(64) [$_{\text{ComP}}$ [$_{\text{DP}}$ *az* ~~[*hogy* … XP]~~] [$_{\text{FocP}}$ [$_{\text{DP}}$ *az* ~~[*hogy* … XP]~~] [$_{\text{FocP}}$ × ✓ [$_{\text{PartP}}$ [$_{\text{DP}}$ *az* ~~[*hogy* … XP]~~] [… [$_{\text{VP}}$ V …

Mari	⟨$_{\textit{TOP}}$	**az-t*	__$_2$⟩	($_{\textit{SADV}}$ *szerencsére)*	[$_{\textit{FOC}}$ *melyik*	*város-on*]$_1$	*akar-t-a*$_3$	⟨$_{\textit{PART}}$	*az-t*	__$_2$⟩	__$_3$	__$_2$,
Mari		it-ACC		fortunately	which	city-SUP	want-PST-3SG.DEF		it-ACC			

[*hogy*	[$_{\textit{PART}}$ *keresztül*]$_4$	*fus-s*	__$_1$	__$_4$	]$_2$?
that	acorss	run-SUBJ.2SG.INDF			

VP-internal or Spec,PartP expletive: 'Which city did you want them to run across?'

As introduced in Section 2.1, Hungarian has multiple focus projections: Foc2P dominating Foc1P, only distinguishable in clauses with multiple foci. In the a. version of (65), focus extraction to Spec,Foc2P can proceed from a clause whose expletive-headed DP-remnant occurs in Spec,Foc1P, i.e. long focus movement lands in front of the expletive. This contrasts with the b. version of (65): Focus extraction to Spec,Foc1P is not possible from a clause whose expletive-headed DP subsequently moves to Spec,Foc2P.

(65) [$_{\text{Foc2P}}$ ✓ [$_{\text{Foc1P}}$ [$_{\text{DP}}$ *az* ~~[*hogy* … XP]~~]… BUT [$_{\text{Foc2P}}$ [$_{\text{DP}}$ *az* ~~[*hogy* … XP]~~] [$_{\text{Foc1P}}$ × …

a. [$_{FOC}$ *CSAK EZ-EN A VÁROS-ON*]$_1$ *akar-t-am*$_3$ [$_{FOC}$ *CSAK AZ-T* __$_2$]$_4$ __$_3$ __$_4$, [*hogy* [$_{PART}$
only this-SUP the city-SUP want-PST-1SG.DEF only it-ACC that
keresztül]$_5$ *fus-s* __$_1$ __$_5$]$_2$
acorss run-SUBJ.2SG.INDF
'The only city I wanted you only to run across was this one.' (Context: Every other city, I wanted you to visit properly.)

b. *[$_{FOC}$ *CSAK AZ-T* __$_2$]$_4$ *akar-t-am*$_3$ [$_{FOC}$ *CSAK EZ-EN A VÁROS-ON*]$_1$ __$_3$ __$_4$, [*hogy* [$_{PART}$
only it-ACC want-PST-1SG.DEF only this-SUP the city-SUP that
keresztül]$_5$ *fus-s* __$_1$ __$_5$]$_2$
acorss run-SUBJ.2SG.INDF

Finally, the expletive positions in (66) indicate that relative movement can proceed from clauses, whose expletive-headed DP-remnant is in Spec,PartP, in Spec,Foc, or in a FocP/ComP-sister topic position.

(66) [$_{\text{RelP}}$ ✓ [$_{\text{ComP}}$ [$_{\text{DP}}$ *az* ~~[*hogy* … XP]~~] [$_{\text{FocP}}$ [$_{\text{DP}}$ *az* ~~[*hogy* … XP]~~] [$_{\text{FocP}}$ [$_{\text{PartP}}$ [$_{\text{DP}}$ *az* ~~[*hogy* … XP]~~] [… [$_{\text{VP}}$ V …

Ez az a város, [$_{REL}$ *amelyik-en*]$_1$ *Mari* ⟨$_{TOP}$ *az-t* __$_2$⟩ ($_{SADV}$ *szerencsére)* ⟨$_{FOC}$ *CSAK AZ-T* __$_2$⟩ *nem*
this that the city which-SUP Mari it.ACC fortunately only it-ACC not
akar-t-a ⟨$_{PART}$ *az-t* __$_2$⟩ __$_2$, [*hogy* [$_{PART}$ *keresztül*]$_3$ *fus-s* __$_1$ __$_3$]$_2$
want-PST-3SG.DEF it-ACC that across run-2SG.SUBJ.INDF
VP-internal or Spec,PartP expletive: 'This is the city which Mari wanted you to run across.'
Focused expletive: 'This is the city which Mari ONLY wanted you to run across.'
Topic expletive: 'This is the city for which the following is true: That you run across it, Mari did not want that.'

The table in (67) summarizes the data in this section; the generalization in (68) summarizes the pattern in (67). Assuming expletives indicate the position of expletive-headed DP-remnants, the general pattern is that subextraction can land in a position c-commanding the DP-remnant but the DP-remnant cannot end up in a position c-commanding the subextracted material. Particle extraction to Spec,PartP is the most constrained as it cannot proceed from any clauses whose DP-remnant occurs in the left-periphery. Focus movement to Spec,FocP can only proceed from clauses whose DP-remnant is in Spec,PartP, or in Spec,Foc1P (the focus subextraction in the last case lands in Spec,Foc2P). Relative movement to Spec,RelP can proceed from clauses whose DP-remnant is either VP-internal, in Spec,PartP, in Spec,FocP, or is a FocP/ComP-sister. Finally, notice that clauses occurring in a FocP/ComP-sister topic position exhibit the same opacity as clauses whose expletive-headed DP occurs in the same FocP/ComP-sister position.

(67)

	Particle extraction to Spec,PartP	Focus/*wh*-extraction	Relative extraction to Spec,RelP
VP-internal clause without expletive	✓	✓	✓
Expletive within Spec,PartP (particle position)	×	✓	✓
Expletive within Spec,Foc1P (focus position)	×	✓ (to Spec,Foc2P)	✓
Expletive within Spec,Foc2P (focus position)	×	× (to Spec,Foc1P)	✓
Expletive within ComP/FocP-sister (topic position)	×	×	✓

(68) **Expletive-mediated position-dependent opacity:** The expletive associated with αP cannot move to a position immediately dominated by a βP, if some XP from αP moved to Spec,γP, and α, β, γ ∈ *fseq* F, and β ≥ γ.

$[_{\gamma P}$ XP [… $[_{VP}$ V … $[_{\beta P}$ × … $[_{\gamma P}$ XP [… $[_{VP}$ V $[_{\alpha P}$ *hogy* … ~~XP~~]]]]]]

So far, (68) has been only tested for biclausal structures: If a clause of origin and a subextracted constituent land in the same left-periphery, the subextracted constituent must c-command the clause. However, consider the triclausal structure in (69): Particle movement from the deepest clause cannot proceed to the matrix Spec,PartP if the expletive associated with the deepest clause occurs in the Spec,FocP of the intermeidate clause. That is, it is not only the case that movement from a clause in Spec,FocP cannot land in the Spec,PartP of the same clause, but it cannot land in the Spec,PartP of any clause (for a similar multi-clausal generalization, see Abels 2008). This general pattern is also depicted in the bracket diagram in (68).

(69) $[_{PartP}$ × [… V [… $[_{FocP}$ $[_{DP}$ *az* ~~[*hogy* … XP]~~] $[_{PartP}$…]]]]

$[_{PART}$ *Keresztül*$]_1$ *kell-ett* $\_\__3$, [*hogy* ⟨$_{FOC}$ **CSAK* *AZ-T* $\_\__2$⟩ *akar-j-am* ⟨$\_\__2$⟩, [*hogy*
across be.needed that only it-ACC want-SUBJ-3SG.DEF that

fus-s-anak *a* *város-on* $\_\__1$ $]_2$ $]_3$
run-3PL.SUBJ.INDF the city-SUP

'It was necessary for me to want to run across the city.'

The unified position-dependent opacity relying on clause-positions (Section 3.2) and expletive positions (this subsection) can be phrased as in (70).

(70) **Unified position-dependent opacity:** If the clause or its expletive occurs in a certain left-periphery position, sub-extraction can only land in a higher left-periphery position.

4 The limitations of previous accounts focusing on internal size

Beyond showing that Hungarian exhibits both size- and position-dependent opacity, the aim of this article is to provide a *unified account* for these two types of selective opacity. Previous accounts of the Williams Cycle and selective opacity in general derived the opacity of constituents from their *internal properties*. In other words, these accounts aimed to derive data exhibiting size-dependent opacity (i.e. data of the style seen in Section 2). Thus, in *delayed substitution* accounts (Williams 2003; Poole 2022; Meadows 2023), it is the maximal projection of the embedded clause that determines when it becomes visible for matrix operations. In Abels' (2012) account, it is the amount and type of left-periphery constituents in the embedded clause that determines which constituents within the embedded clause are accessible while obeying Relativized Minimality. In Keine's (2016; 2019; 2020) account, it is the feature content of the maximal projection of the embedded clause that allows or blocks subextraction. Essentially, these accounts do not pay attention to the syntactic position of the embedded clause in the matrix clause.

If one wants to keep the same mechanism and reasoning of these previous accounts while deriving position-dependent opacity (introduced in Section 3), the most natural option is to connect the position of a constituent to its internal structure: If a constituent wants to show up in a certain position, it must be of a certain size. However, I demonstrate in this section that this way of reducing position-dependent in (71) opacity to size-dependent opacity in (72) does not cover all cases because size and position do not always go hand in hand.[12]

(71) **Position-dependent opacity**: αP or the expletive-headed DP associated with αP cannot move to a position immediately dominated by a βP, if some XP from αP moved to Spec,γP, and α, β, γ ∈ *fseq* F, and β ≥ γ.

(72) **Size-dependent opacity/Strict Williams Cycle:** Movement to Spec,βP cannot proceed across a dominating αP, where α ≥ β in the functional sequence.

Cable (2007; 2010b; 2010a) claims that *wh*-phrases do not belong to various categories (DPs, PPs, AdvPs etc.) but *wh*-movement uniformly targets QPs. In other words, the *position* of *wh*-words in Spec,CP is connected to their *size* because all constituents landing in a certain $\overline{\text{A}}$-position must be of a certain size/category. This intuition of connecting size and position is generalized in (73), which claims that embedded clauses occurring higher in the matrix are larger. For example, if an embedded clause is contained in matrix Spec,ComP, this embedded clause must be at least a ComP′. By (72), movement to Spec,FocP is blocked from this ComP′ clause because Com>Foc in *fseq*. On the other hand, if the clausal expletive occurs in Spec,PartP, the clause itself is possibly a PartP, from which focus extraction can proceed to Spec,FocP because Foc>Part in *fseq*. Thus, in theory, position-dependent opacity in (71) is redundant if it can be

[12]Müller (2014a; 2014b) proposes a framework that determines the grammaticality of movement operations on the basis of movement paths arising from the crossed maximal projections. Since the movement path of cross-linguistic movement operations has a section in the matrix clause, this account considers clause positions. Although my proposal in Section 5 is similar to Müller's in certain respects, Müller only considers sub-extraction clauses that are complements to V. Furthermore, my analysis in Section 5.2 provides a principled explanation why domains that can be escaped successive cyclicly do not exhibit size- and position-dependent opacity. Since Müller's analysis of size-dependent opacity relies on the idea that movement must stop in every phrase, his analysis is not able to connect successive cyclicity and the lack of size-dependent opacity.

traced back to the combination of size-dependent opacity in (72) and the position-size correlation in (73). That is, at first sight, it seems that frameworks deriving size-dependent opacity could use the position-size correlation to derive position-depdent opacity.

(73) **The position-size correlation:** If an embedded clause or its expletive-headed DP is the sister of matrix βP, this clause must be αP such that α ≥ β in *fseq*.

Crucially, the position-size correlation in (73) predicts that position- and size-based opacity always go hand in hand. If a constituent originating in the embedded clause can move to the matrix position X across another constituent in the Y position of the embedded clause, movement to matrix X should not be blocked if the embedded clause occurs in the Y position of the matrix clause. For example, clauses with particles (PartPs) do are transparent to long focus movement as in (33), and *wh-* and focus movement is possible from clauses whose expletive occurs in Spec,PartP as in (64).

So far, topics and quantifiers have been left out from the discussion. Topics and quantifiers are special in that they always precede foci and particles in the left-periphery (see Section 2.1), but Egressy (2025) describes that neither topics nor quantifiers in the embedded clause block the long movement of particles, foci, and relative pronouns (partially also described by Koopman and Szabolcsi 2000). For example, (74) indicates that topics do not block focus extraction from embedded clauses. As long as topics do not have a dedicated RefP/TopP projection such that Top/Ref>Foc in *fseq* (as in Brody and Szabolcsi 2003) but topics are able adjoin to maximal Foc1P projections in (74) in a way that the embedded clause is a Foc1P, the grammaticality of focus sub-extraction to Spec,Foc2P is compatible with (73).

(74) $[_{\text{Foc2P}}$ ✓ … $[_{\text{Foc1P}}$ VP … [topic [… XP …]]]]

[*FOC* *CSAK*	*A*	*VÁROS-ON*]$_1$	*akar-t-am*$_2$	__$_2$, [	*hogy*	[*TOP* *holnap*]	*keresztül*$_3$	*fus-s-anak* __$_1$ __$_3$]
only	the	city-SUP	want-PST-1SG.INDF		that	tomorrow	across	run-SUBJ-3PL.INDF

'As for tomorrow, I wanted them to run across ONLY THE CITY .'

However, topic-clauses and clauses whose expletive-headed DPs are in a topic position do exhibit position-dependent opacity, as the examples of Sections 3.2 and 3.4 indicated. More specifically, Section 3.4 has shown that clauses whose expletive occurs in a topic position are more opaque than clauses whose expletive occurs in a focus projection: focusing can only proceed from the latter because expletives in topic position are higher in the left-periphery than expletives in focus position. That is, if we wanted to derive this kind of position-dependent opacity from internal size in the spirit of (73), we should assume that there is a projection within topic clauses and clauses whose expletive is in a topic position which clauses whose expletive is in a focus position do not have. For example, while clauses whose expletive is in a Spec,Foc1P are Foc1Ps, clauses which are topics or whose expletive occur in a topic position are TopPs/RefPs such that Top/Ref>Foc in the *fseq*. However, we run into a contradiction: If there was a TopP/RefP to host topics, and this TopP/RefP dominated FocPs in the clausal extended projection, we would expect that (74) was degraded because size-dependent opacity would not allow movement from TopP to Spec,FocP if Top/Ref>Foc in the *fseq*. In short, we cannot use (73) and derive the opacity of topic clauses and clauses whose expletives are in topic positions from their internal properties or size because topics do not contribute to the size-dependent opacity of clauses. The same problem arises with quantifiers: While quantifiers, which occur above foci and particles in the left-periphery, do not contribute to size-dependent opacity (Egressy 2025), clauses whose expletives are in a quantifier position exhibit position-dependent opacity (e.g. they are opaque to focusing; Egressy 2023:Example 136).

In sum, the attempt to reduce position-dependent opacity in (71) to a combination position-dependent opacity (72) and the position-size correlation in (73) is unsuccessful. No account focusing on internal size only can derive position-dependent opacity in Hungarian. For this reason, my proposal in Section 5 basically turns around (73) by claiming that larger clauses must be in a higher position in order to be transparent, i.e. size requires position rather than the way around.

5 Proposal

5.1 Move Along Increasing *fseq*-indices (MAIF)

As argued in Section 4, the opacity of Hungarian embedded clauses cannot be reduced solely to their *internal structural size*, that is, accounts focusing on size-dependent opacity are insufficient. In another family of accounts, the transparency of embedded clauses is influenced by their *syntactic environment* in the matrix clause. For example, Rackowski and Richards (2005), Van Urk and Richards (2015), and Den Dikken (2018) claim that the embedded clause becomes transparent if the matrix verb agrees with it. However, since these accounts operate with absolute terms of phasality (i.e. it is always the *v* that agrees with CP), they are unable to capture opacity-differences arising from internal size. Thus, we need an account that pays attention to both internal size of embedded clauses and their position in the matrix clause. This section introduces Move Along increasing Functional Sequence Indices, which is a size-sensitive alternative to PIC.

Syntactic operations have been found to be constrained by locality domains, which have been called *phases* in the last decades. In standard phase theory (Chomsky 2000, 2001), the Phase-Impenetrability Condition (PIC) in (75) specifies the positions which are available from outside a phase. In Chomsky's original work, phasehood is *absolute* in the sense that it is rigidly associated with specific phrase labels. For example, the phrase headed by C is a phase and always a phase. Therefore, whenever the building of a CP is complete, no material c-commanded by the C head (i.e. nothing in the *domain* of C) can leave the CP. Further categories associated with phasehood including *v* and D (e.g. Chomsky 2000; Svenonius 2004) are handled identically. Fundamentally, the (un)availability of the material c-commanded by the phase-head is not dependent on the landing site of the movement relative to the size and position of the phase. In a similar way, if something moves to the *edge* position of a phase, it can move to any height in next lowest phase.[13]

[13]While certain theories claim that only $\overline{\text{A}}$-movement cannot proceed from Spec,CP because of the Ban on Improper movement (Chomsky 1973), the Williams Cycle literature has found that the A vs. $\overline{\text{A}}$-distinction is not sufficient to derive most Williams Cycle effects. For example, Hungarian has *three* movement operations each exhibiting different locality constraints (but also see Meadows 2023).

(75) **Phase-Impenetrability Condition (PIC)**: The domain of H is not accessible to operations outside HP; only H and its edge are accessible to such operations. (Chomsky 2000:p.13).

$[\times \ldots [_{\text{HP phase}} \checkmark\ [\text{ H } [\ldots \text{XP} \ldots]]]]$ $\qquad$ $[\checkmark \ldots [_{\text{HP}\mathit{phase}}\ \text{XP} [\text{ H } [\ldots]]]]$

In short, default phase theory relying on the PIC in (75) and phasehood rigidly associated with particular categories is *size- and position-insensitive.* In contrast, Sections 2 and 3 argued that the opacity of the various clauses in Hungarian is sensitive both to *size* and *position.* Therefore, the *absolute* definitions of standard phasehood insensitive to size and position are not sufficient while analyzing *selective* opacity in Hungarian.

To capture the facts of size- and position-dependent opacity while also keeping successive cyclicity in the picture, I replace the classic PIC in (75) with the constraint *Move Along Increasing fseq-indices* (MAIF) in (76). The idea that syntactic movement needs to be internal to a (clausal) domain goes back to Chomsky (1977: p. 85), who claims only clause-internal movement is universal; cross-clausal movement arises independently, via language-specific rules. Furthermore, Bošković (2014) claims that syntactic locality (for him, phasehood) needs to be constrained by the highest projections of extended projections, rather than projections of a certain category (e.g. CP or DP). (Extended projections are tokens of functional sequences.) MAIF unifies the intuitions by Chomsky (1977) and (Bošković 2014): Syntactic movement universally proceeds within extended projections; movement across extended projections needs to be turned into steps that are internal to external projections. The technical implementation of this idea is the following: Along the path of the movement, it must hold that the crossed heads belong to the same *fseq* and that their *fseq*-indices increase.[14]

(76) **Move Along Increasing *fseq*-indices (MAIF):** XP can move from A to B iff the heads crossed form a well-formed functional sequence fragment.

The term *crossing* is defined as follows:

(77) **Crossing:** For any two heads α and β both c-commanding the original position of XP and both c-commanded by the landing site of XP, α is higher than β in the sequence of crossed heads by XP iff α asymmetrically c-commands β.

The term *well-formed functional sequence fragment* is defined in (78). In Grimshaw's 2000 terms, "α higher in *fseq* β" means "α has a higher *fseq*-index than β". Thus, if *fseq* is V<*v*<T<C, the sequences V-*v*-T and *v*-T-C are well-formed *fseq*-fragments but *V-***v*-C** and ***C-V**-*v*-T are not well-formed because of the missing T head and the "index-drop" between C and V, respectively. In order to make the sequences of crossed heads comparable to *fseq*s, I will unconventionally write *fseq*s starting with the lower heads. The points of crash will be marked with boldface.

[14] The idea of relating movement to a well-formed paths goes back at least to Kayne (1984); constraining movement paths in terms of category labels in *fseq* is reminiscent of the proposal in Müller (2014a; 2014b).

(78) **Well-formed functional sequence fragment:** It holds for every α-β two-head sub-sequence of the sequence of heads crossed by XP that

(i) α and β belong to the same functional sequence,

(ii) α is higher than β in this functional sequence, and

(iii) there is no γ such that α>γ>β in this functional sequence.

The output of MAIF is similar to the spirit of Bošković (2014): Locality-boundaries coincide with the boundaries between extended projections.[15] In (79), CP′ is a locality-boundary: XP cannot be subextracted from CP′ in one step to a position c-commanding the matrix V because this movement step would proceed along the *V′-*v*′-T′-**C′-V** sequence, which would violate (76) as V<C in *fseq*. TP′ is not a domain-boundary as long as it is selected by a C′, which is part of the same *fseq* as in (79): XP can move along the V′-*v*′-T′-C′ sequence to Spec,CP′. Nevertheless, the embedded TP′ becomes locality-boundary in (80) once it is directly selected by a matrix V, which starts another *fseq*. In this case, an XP subextracted from TP′ would have to move along the *V′-*v*′-**T′-V** sequence.

(79) $[\ \times\ [_{\text{VP}}\ \text{V}\ [_{\text{CP}'}\ \text{C}'\ [_{\text{TP}'}\ \text{T}'\ [_{v\text{P}'}\ v'\ [_{\text{VP}'}\ \text{V}'\ \cancel{\text{XP}}]]]]]]$

(80) $[\ \times\ [_{\text{VP}}\ \text{V}\ [_{\text{TP}'}\ \text{T}'\ [_{v\text{P}'}\ v'\ [_{\text{VP}'}\ \text{V}'\ \cancel{\text{XP}}\ ]]]]]$

However, while Bošković (2014) does not depart from the original PIC in (75) in that he keeps the concept of phases in the theory, MAIF in (76) is similar to recent work by Halpert and Zeijlstra (2024) and Keine and Zeijlstra (2025) in that it does not make reference to phases. That is, in the context of the theory of movement presented here, the concept of phases can be dropped or loosened (e.g. to *Read only* suggested by Agarwal (2022; to appearb; to appeara)). At the same time, abandoning phases would raise questions on phase-related operations not necessarily involving movement e.g. ellipsis, Case, and Agree. Due to space limitations, I leave it to further research to decide whether the current MAIF-based theory can be a pathway to loosen/drop phases from syntactic theories.

5.2 Successive cyclicity as a way around size- and position-dependent opacity

Following the intuition of Chomsky (1977: p. 85), I propose that there are various strategies to make cross-clausal movement grammatical; these are various *escape strategies*. In the context of MAIF, an escape strategy turns cross-clausal movement into movement steps which are internal to clauses/extended projections. The available escape strategy is the property of individual projections/categories in individual languages. This subsection discusses successive cyclicity, which is an escape strategy available, for example, for P'urhepecha or Mongolian CPs. Section 5.4 describes the movement of clauses as an alternative escape strategy available for the clausal projections of Hungarian or German CPs. If there is no escape strategy available for a certain projection in a certain language, cross-clausal movement is

[15] Bošković (2014) assumes that the clausal spine contains two *fseq*s, a *v*-*fseq* and a C-*fseq*. For now, I assume that clausal spines are a single *fseq*.

unavailable. For example, the CP projection in Tsez (Polinsky and Potsdam 2001) cannot be escaped via either of the strtegies.

Successive cyclic movement is one way to avoid locality violations either under the head-based MAIF definition in (76) or the phrase-based PIC defnition in (75). Essentially, both PIC and MAIF claim that within some locality domain αP, the movement of constituents c-commanded by α is banned whereas constituents asymmetrically c-commanding α (i.e. α's specifier(s)) are accessible from outside αP. In other words, the *edge* status of specifier(s) of locality domains is preserved under MAIF in (76): XP in the Spec,αP of some locality domain αP c-commands every head in α, nothing in αP constrains further movement of XP.[16] For example, while subextraction is impossible in one step from CPs and TPs in (79) and (80), respectively, successive cyclicity makes cross-clausal movement MAIF-compliant. Assuming that *fseq* is V<*v*<Part<C for (81), if CP is complement to the embedded V′, XP can move to the embedded Spec,CP, as it moves along the V′-*v*′-T′-C′ sequence in the first movement step in (81). In the second step, the first head crossed is the matrix V, after which it only crosses matrix heads, that is, movement from Spec,CP is internal to the matrix clause as far as MAIF is concerned. Thus, unless operators are criterially frozen in Spec,CP, XP can leave Spec,CP and move into a position within the matrix clause. In the same way, subextraction from TP′ can proceed in two steps via Spec,TP′ in (82): The first step proceeds along the V′-*v*′-T′ sequence and the matrix V is the first head to cross in the second step.

(81) $[\ \checkmark\ [_{\text{VP}}\ \text{V}\ [_{\text{CP}'}\ \text{XP}\ [\ \text{C}'\ [_{\text{TP}'}\ \text{T}'\ [_{v\text{P}'}\ v'\ [_{\text{VP}'}\ \text{V}'\ \cancel{\text{XP}}]]]]]]]$ (82) $[\ \checkmark\ [_{\text{VP}}\ \text{V}\ [_{\text{TP}'}\ \text{XP}\ [\ \text{T}'\ [_{v\text{P}'}\ v'\ [_{\text{VP}'}\ \text{V}'\ \cancel{\text{XP}}\ ]]]]]]$

A certain αP constituent in a language L may have one, multiple, or no edge positions, via which operator movement can proceed successive cyclically. In the following subsections, the core of my proposal will be that the size- and position-dependent opacity of Hungarian clausal domains arises because MAIF-violations cannot be avoided via successive cyclic movement in the absence of the necessary edge positions in the clausal domain.[17] In contrast, if cross-clausal movement *can* proceed via the edge of a clause type of a language, MAIF on its own does not pose any restrictions on the landing height. That is, the MAIF-based account predicts that if no independent factors constrain movement from a clause, size- and position-dependent opacity do not arise under successive cyclicity.

For example, successive cyclicity enables *hyperraising* from CPs; see Zyman's (2017; 2018) discussion on P'urhepecha CPs and Fong's discussion (2019) on Mongolian CPs (for more discussion, see Halpert and Zeijlstra 2024). From CPs, hyperraising to Spec,*v*P or Spec,TP violates the Williams Cycle or Size-dependent opacity repeated in (83) because C>T>*v* in *fseq*. However, if hyperraising proceeds successive cyclically, these violations of the Williams Cycle are

[16]The MAIF definition in (76) can be re-phrased in terms of dominating phrases: The category-labels in the sequence of phrases that dominate the launch site A but not the landing site B of XP must form a well-formed *fseq*-fragment. In the main text, I stick to the simpler head-based definition because the phrase-based definition requires a distinction between "dominated by every segment of" vs. "dominated by some segments of" (as in e.g. Chomsky 1986, 1995) to exclude αP from the set of crossed dominating phrases for movement launching in Spec,αP (i.e. to preserve the *edge* status of Spec,αP).

[17]Successive cyclicity can be posited Hungarian PPs and DPs. According to Dékány and Hegedűs (2015), the movement of certain Hungarian P-complements can proceed successive cyclically, which can be followed by the remnant movement of the postposition to a Spec,PartP position. Furthermore, possessive DPs have an edge position available exclusively for possessor DPs, where possessor DP move to get dative case (e.g. Szabolcsi 1983; Alberti and Farkas 2018; den Dikken and Dékány 2018). Essentially, clauses cannot escape DPs via their edge, as I will argue in Section 5.7.

obviated (i.e they do not lead to ungrammaticality).

(83) **Size-dependent opacity/Strict Williams Cycle:** Movement to Spec,βP cannot proceed across a dominating αP, where α ≥ β in the functional sequence.

The P'urhepecha example in (84) is argued to exhibit hyperraising to object in a way that the embedded subject passes through Spec,CP and lands in Spec,*v*P (Zyman 2017:p.4). That the movement of *Xumoni* lands in Spec,*v*P position in (84) is shown by its position between the matrix verb and a PP modifying the matrix VP. Again, hyerraising is MAIF-compliant if it stops in Spec,CP because the subject moving from Spec,TP to Spec,CP only crosses C, the second movement step proceeds along the V-*v* sequence. That the CP has *one* edge via which moving constituents can escape is demonstrated by (85) (Zyman 2017:p.4): If *wh*-movement uses the edge of CP to escape, hyperraising to object does not have a second edge position via which it can proceed, that is, *wh*-movement and hyperraising cannot proceed from the same CP. In contrast, Mongolian CPs have multiple edge positions, so multiple movement operations can proceed from a single clause (Fong 2019).

(84) *Emilia uekasïndi*[$_{vP}$ *Xumoni*$_1$ *mintsitani jingoni* [$_{CP}$ __$_1$ *eska* __$_1$ *jaruataaka pauani*]].
Emily want.IND3 Xumo.ACC heart.ACC with that help..SBJV tomorrow
'Emily wants Xumo with all her heart to help her tomorrow.'

(85) *¿Ambe*$_1$ *uetarinchasïnga*[$_{vP}$ ⟨*??Emiliani*⟩$_2$ [$_{CP}$ __$_2$ __$_1$ *eska* ⟨*Emilia*⟩$_2$ *k'uanatsintaaka*_$_1$]]*?*
what need.INT.pS Emily.ACC that Emily buy.SBJV
'What do they need Emily to buy?'

In sum, if an extended projection has at least one specifier where operators are not criterially frozen, successive cyclic movement can proceed from within this projection. MAIF predicts that movement from an edge position can land on any height in the next lowest extended projection.[18]

[18] Apart from analyses arguing for *wh*-cluster-formation (Grewendorf 2001), the Bulgarian CP has been argued to have multiple specifiers/edges (Richards 1997, 2004; Zompì 2024). Furthermore, it has been found that Bulgarian has remnant-movement configurations like (i), where the subextracted *wh*-phrase ends up in the embedded Spec,CP and the *wh*-remnant containing it in the matrix Spec,CP (Richards 2004:p.459). (i) can get a MAIF-compliant derivation: The outer *wh*-phrase YP first moves to the embedded edge position Spec,CP, which is within-clausal movement in the embedded clause. After this, the inner *wh*-phrase XP can *tuck-in* to a lower Spec,CP in a MAIF compliant way as it only crosses heads within YP, i.e. this tucking-in step is MAIF-complaint. This can be followed by the movement of the remnant YP to the matrix Spec,CP. In sum, YP does not need to have an edge if CP has multiple edges: successive cyclic movement from within a YP in a Spec,CP is MAIF-compliant to another Spec,CP.

(i) [✓ [$_{VP}$ V [$_{CP'}$ [$_{YP}$ … ~~XP~~ …] XP [C′ [$_{TP'}$ T′ [$_{vP'}$ v' [$_{VP'}$ V′ ~~YP~~]]]]]]]

[$_{YP}$ *Kolko studenti* __$_1$]$_2$ *se opitvaš da razereš* [$_{CP}$ __$_2$ [$_{XP}$ *ot koi strani*]$_1$ *e ubil Ivan* __$_2$*?*
how.many students REFL you.try to understand from which countries AUX killed Ivan
'[$_{YP}$ how many students __$_1$]$_2$ are you trying to understand [$_{CP}$ __$_2$ [$_{XP}$ from which countries]$_1$ Ivan killed __$_2$?'

5.3 Hungarian *hogy*-clauses lack edges

In this section, I propose that the Hungarian clausal left-periphery does not have edge positions, i.e. successive cyclic movement is impossible from Hungarian finite clauses. The specifiers introduced for Hungarian are Spec,RelP for relative pronouns, Spec,ComP for sentential adverbs, Spec,Foc1P and Spec,Foc2P for foci, and Spec, PartP for particles. Following Rizzi's analysis for Italian (Rizzi 1997), I assume that these left-periphery positions are criterial positions and movement to these positions bring about *criterial freezing*: Constituents landing in these specifiers cannot move further, that is, these specifiers cannot be used as escape hatches. I assume, moreover, that sModP does not have a specifier.

That Hungarian left-periphery specifiers cannot be used as escape hatches is supported by the absence of visible reflexes via these positions. For example, Hungarian focus movement triggers a reflex at its landing site: As shown in Section 2.1, whenever Spec, FocP is filled, the verb moves in front of its particle in Spec,PartP to the Foc head. Therefore, we would expect verb-particle inversion in embedded clauses if long-moving constituents used the embedded Spec,FocP as an escape hatch. However, this is not the case: Under long focus movement in (86), verb-particle inversion occurs only in the matrix clause with *el*, whereas the particle *keresztül* in the embedded clause stays in front of its verb (Horvath 1981, 1986; Richards 1997).

(86) [$_{\text{FocP}}$ ✓ Foc+✓ [$_{\text{PartP}}$ particle … [$_{\text{VP}}$ V [Foc+× [$_{\text{PartP}}$ particle … [$_{\text{VP}}$ V [… XP …]]]]

[$_{\text{FOC}}$ *CSAK A VÁROS-ON*]$_2$ *ér-t-em*$_1$ [$_{\text{PART}}$ *el*] __$_1$ *pro,* [*hogy* [$_{\text{PART}}$ *keresztül*]$_3$
only the city-SUP achieve-PST-1SG.INDF PART pro.ACC that across

fus-s-anak __$_1$ __$_3$]
run-SUBJ-3PL.INDF

'I accomplished that they would run across the ONLY THE CITY.'

The verb-particle inversion is also absent in the embedded clause of (36) suggesting that long relative movement does not have an intermediate stop in Spec,FocP′, either.

In the last subsection, P'urhepecha CPs were demonstrated to have one edge position, which allowed only one sub-extraction per CP. This contrasts with the Hungarian example in (87): the movement of the relative pronoun and the particle can proceed from the same clause. If the embedded clause had a *single edge* position, the successive cyclic movement of two operators would be impossible. (This is not an argument against multiple edge positions.)

(87) [$_{\text{RelP}}$ ✓ … [$_{\text{PartP}}$ ✓ … [$_{\text{VP}}$ V [… V [… XP YP …]]]]]

A város, [$_{\text{REL}}$ *amelyik-en*]$_1$ *ma* [$_{\text{PART}}$ *keresztül*]$_2$ *akar-t-ad,* [*hogy fus-s-anak* __$_1$ __$_2$]
the city which-SUP today across want-PST-2SG.INDF that run-SUBJ-3PL.INDF
'The city which today you wanted them to run across.'

Thus, my conclusion is similar to that of Den Dikken (2009; 2018): No phrases can be subextracted from embedded RelPs, ComPs, FocPs, PartPs, and sModPs in a successive cyclic manner.[19] In general, the absence of edges is not a language-wide parameter, e.g. the possiblity of via Spec,PPs in Hungarian is briefly discussed in Footnote 17. Rather, the absence of edges in the Hungarian clausal domain is simply the absence of a projection not exhibiting criterial freezing.

5.4 The movement of clauses as an alternative strategy

In the absence of successive cyclic long movement via left-periphery edges, I propose that cross-clausal movement in Hungarian remains MAIF-compliant via an alternative strategy: the adjunction of embedded clauses with an equal-sized matrix position.

This section focuses on expletive-less constructions in which the finite clause in Hungarian is the complement of the matrix V. As discussed in Section 3.1, if the expletive-less finite clause is internal to the head-final VP, it can adjoin to any head-initial matrix projection to avoid the violation of the Final-over-Final constraint. The general intuition behind the analysis is that while embedded clauses leaving the VP can adjoin at various heights in the matrix clause, subsequent subextraction is only possible with certain landing sites. (In the spirit of Privoznov (2021), adjuncts are not necessarily islands.) That is, my proposal is different from the proposals of Endo and Haegeman (2019) and Egressy (to appear), which assume that embedded clauses always merge with equal-sized matrix projections. My proposal is more similar to that of Davison (2009), who claims that the size-relation between clauses may be both symmetric and asymmetric but symmetric structures exhibit different properties than asymmetric ones.

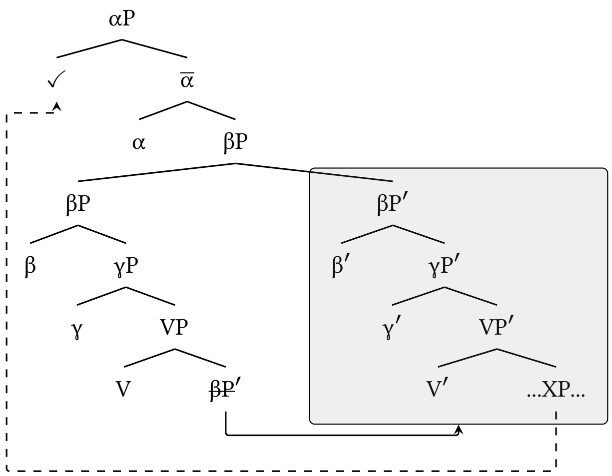
In the absence of successive cyclic movement from Hungarian clauses, MAIF-compliant cross-clausal movement is only possible if the embedded clause adjoins to an *equal-sized* matrix projection. (MAIF is repeated in (88).) This is exemplified by the βP-βP′ adjunction in (89). The movement of the embedded βP′ is MAIF compliant within the matrix clause as it only crosses the heads of one extended projection. In (89), given that the clausal *fseq* is V<γ<β<α, the sequence V-γ-β, along which the embedded clause moves, is MAIF-compliant. Supposing the structure-building continues in the matrix clause even after the adjunction of the embedded clause as in (89), the sequence of heads c-commanding the XP in the adjoined embedded clause is the following: V′-γ′-β′-α . Thus, XP can move to Spec,αP because the *fseq*-indices of the heads keep increasing between the two copies of XP. In other words, the subextraction of XP to Spec,αP is compatible with MAIF in (88) because the embedded *fseq* finds a continuation in the matrix *fseq*.

[19] While the early literature on Hungarian proposed successive cyclic movement via clause edges, these phenomena were re-analyzed in the subsequent literature. Firstly, if a *wh*-question is formed about the subject of a *hogy*-clause, the *wh*-word occurring at the edge of the matrix clause receives accusative case; roughly the following manner: *Who*(-ACC)$_1$ you.think* [*that* __$_1$ *saw John*]*?* for 'Who do you think saw John?'. Among others, Chomsky (1981) and Massam (1985) analyze this as evidence for a successive cyclic stop at the edge of the embedded clause, where the accusative case is assigned to the moving *wh*-phrase. However, such examples have been re-analyzed as *prolepsis*, i.e. the *wh*-phrases are generated in the matrix clause and bind a silent pronominal in the embedded clause (e.g. Jánosi 2013; den Dikken 2018; Poole 2022). Secondly, Marácz (1989) claims that Hungarian does not allow multiple operator movement from embedded clauses as there is just a single edge position via which long-operator movement can proceed. However, (87) is a counterexample: the relative pronoun and the particle move out from the same clause without forming a cluster. Finally, partial *wh*-movement in Hungarian can take the following shape: *What-ACC you.think* [*that who-ACC$_1$ John saw* __$_1$]*?* for 'Who do you think John saw?'. While Marácz (Marácz 1987) proposes that the *wh*-phrase in the embedded clause is a pronounced link in a chain exhibiting successive cyclicity, Horvath (1997) argues convincingly that partial *wh*-movement does not involve genuine cross-clausal movement, i.e. partial and full *wh*-movement are two fundamentally different constructions.

The terminology of Bošković (2014), this apparent case of cross-clausal movement remains within a single extended projection because the highest projection of the embedded clause ceases to be the maximal projection in the extended projection after the adjunction of the embedded clause.

(88) **Move Along Increasing *fseq*-indices (MAIF):** XP can move from A to B iff the heads crossed form a well-formed functional sequence fragment.

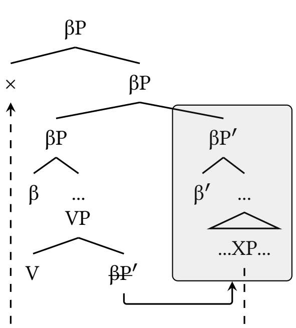
(89)



Notice that as long as the embedded clause is VP-internal, the next highest head above β′ is V, that is, β′ tops a sequence of heads constituting a well-formed *fseq*-fragment. Thus, under MAIF in (88), the VP-internal βP′ is a barrier for movement: No XP can leave it as that movement operation would cross β′ and then V in one step, which does not follow the increase-in-*fseq*-indices requirement.

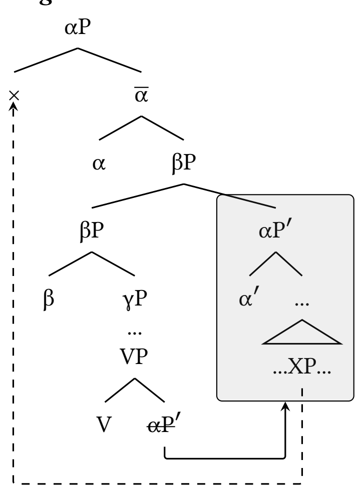
Movement operations can only land in specifiers that c-command the adjunction site. Thus, subextraction to Spec,αP is possible in (89) but subextraction to Spec,γP is impossible because while the matrix γP is built, embedded βP′ is still internal to the matrix VP. Moreover, within a certain projection, adjunction is generally assumed to happen after obligatory feature-checking movement operations (see e.g. Hunter 2010, 2015). Hence, the projection emerging above the adjunction site of the embedded clause, βP in (90) cannot have a specifier above this landing site. Consequently, no Spec,βP c-commanding the both clauses is available if the embedded clause is a βP.

(90)

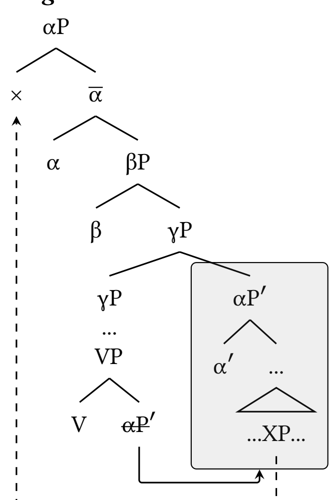


Although embedded clauses are capable of adjoining to non-equal-sized matrix projections because these are also movement steps internal to the matrix clause, these asymmetric configurations do not make subextraction from embedded clauses possible. Firstly, MAIF in (88) rules out subextraction from embedded clauses that have been adjoined to a matrix projection that is smaller than their size. In (91), a larger αP′ adjoins to a smaller βP. (*fseq* = V<γ<β<α.) Here XP cannot be subextracted to Spec,αP even after its adjunction because XP would move along the ill-formed *V′-γ′-β′-**α′-α** sequence. Specifically, since two heads have the α label, this is not a well-formed *fseq*-fragment, i.e. MAIF is violated as the constant increase in *fseq*-indices does not hold. In (92), the embedded αP′ merges with an even smaller matrix γP. In this case, subextraction from the embedded clause to matrix Spec,αP proceeds along the *V′ -γ′-β′-**α′-β**-α sequence, which includes a decrease in the *fseq*-indices and hence violates MAIF. To put it another way, since the αP′ remains a maximal projection in the extended projection in (91) and (92), no movement can leave it. Secondly, MAIF rules out subextraction in *smaller-to-larger* configurations, too. In (93), γP′ adjoins to a larger βP. XP cannot move to the matrix Spec,αP because the sequence of heads crossed would be *V′-**γ′-α**. This sequence is not a well-formed *fseq*-fragment because γ′ and α are not adjacent in *fseq*; β is missing. Intuitively, the continuity of the extended projection is broken.

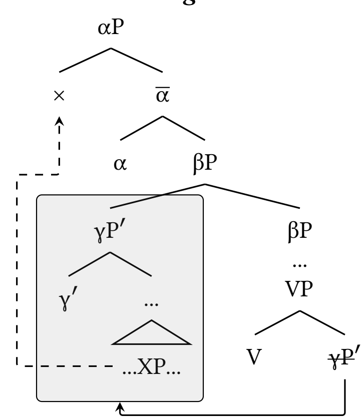
(91) ***Larger-to-smaller***



(92) ***Larger-to-much-smaller***



(93) ***Smaller-to-larger***



In sum, Hungarian embedded clauses become transparent to subextraction after adjoining to an *equal-sized* matrix projection. The embedded clause's adjunction to an equal-sized matrix projection follows the more general idea of "unlocking by clause-*external* operations" (e.g. Rackowski and Richards 2005; Van Urk and Richards 2015). However, the unlocking mehtod presented here is size-sensitive: Every clause gets unlocked by adjoining to its own size in the matrix clause.

5.5 Deriving size-dependent opacity in Hungarian and beyond

Embedded clauses only become transparent to subextraction if they adjoin to an equal-sized matrix projection, and movement can land only land in specifiers c-commanding the landing site. It directly follows that smaller embedded clauses can adjoin to an equal-sized matrix projection lower whereas larger embedded clauses can adjoin to an equal-sized matrix projection higher. Lower matrix specifiers will only c-command the adjunction sites of smaller embedded clauses, so fewer types of sub-extraction will be possible from larger clauses and lower matrix clause positions will be harder to reach.

In formal terms, an embedded βP′ is only transparent to MAIF-complaint sub-extraction if it merges with a matrix βP. Notice that below this point, every matrix projection is a γP such that γ < β in *fseq*; i.e. the adjunction of the embedded βP′ to these γPs would result in opaque *larger-to-smaller* configurations like (91) and (92). Once the βP′ merges with the equal-sized βP, every matrix specifier c-commanding this adjunction site is an instance Spec,αP such that β <α in *fseq*. Consequently, the generalization on size-dependent opacity (repeated in (94)) is derived if a language uses the movement of embedded clause to enable MAIF-compliant cross-clausal movement.

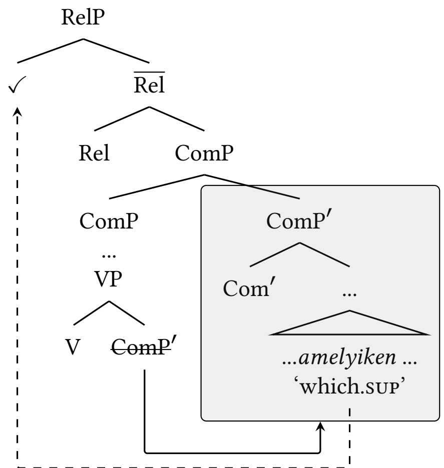
(94) **Size-dependent opacity/Strict Williams Cycle:** Movement to Spec,βP cannot proceed across a dominating αP, where α ≥ β in the functional sequence.

The Hungarian clausal *fseq* established in Section 2 is the following:

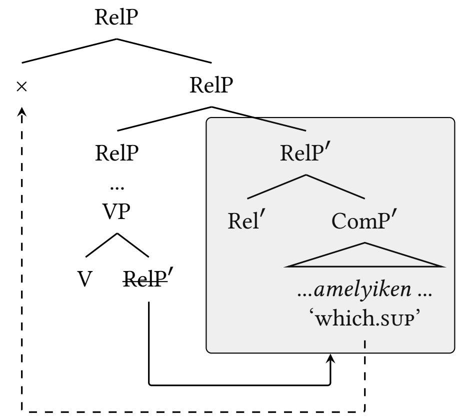
(95) **Clausal *fseq***: V<sMod<Part<Foc1<Foc2<Com<Rel
(in clauses with one focus, there is no Foc1-Foc2 distinction)

The generalization in (94) was drawn on the basis of the following movement types: Relative movement landing in Spec,RelP, focus movement landing in Spec,Foc2P and Spec,Foc1P, and particle movement landing in Spec,PartP. Relative movement can proceed from ComPs and smaller clauses but not from RelPs. Since RelP is the highest projection of the extended projection, Spec,RelP c-commands embedded clauses that are adjoined the matrix ComP or lower. The tree diagram in (96) demonstrates this with a ComP-ComP′ adjunction. ComP′ first moves along the well-formed V-sMod-Part-Foc-Com sequence and adjoins to the equal-sized matrix ComP. Subsequently, the relative pronoun moves along the well-formed sequence V′-sMod′-Part′-Foc′-Com′-Rel. That is, this derivation of cross-clausal relative movement is MAIF-compliant. Relative extraction from embedded clauses smaller than ComP will only differ from (96) in the height of the adjunction because the adjunction of smaller clauses to equal-sized matrix projections happens lower (e.g. FocP′ with FocP, PartP′ with PartP, and sModP′ with sModP). The impossibility of relative movement from RelPs in (97) follows from that no specifier projects above the adjunction site of embedded clauses, i.e. no Spec,RelP projects above the adjunction site of RelP′ just as we saw in (90). That is, the absolute islandhood of Hungarian relative clauses is derived because no matrix specifier c-commands RelPs adjoined to matrix RelPs.

(96)

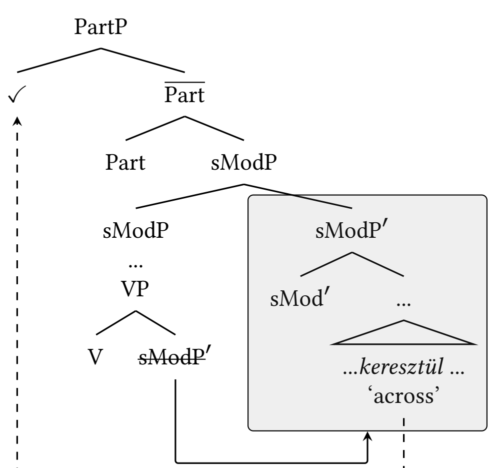


(97)

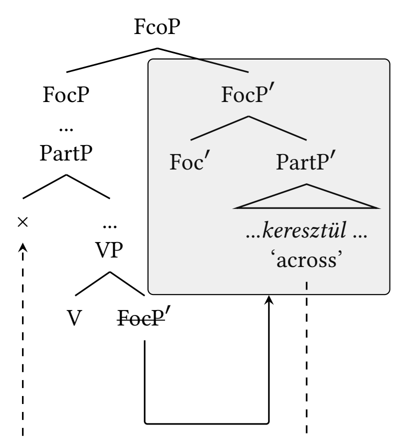


At the other extreme, the subextraction of particles is only possible from the smallest subjunctive clauses identified here as sModPs. In (98), sModP′ adjoins to the matrix sModP in a way that the shared Spec,PartP c-commands the embedded clause. (The movement of the clause happens within the matrix clause, i.e. in a MAIF-compliant way.) After this, particles move along the well-formed V′-sMod′-Part sequence. Just to avoid Final-over-Final violations, larger clauses like FocPs and ComPs can also adjoin to the matrix sModP. However, since the emerging structures exemplify the *larger-to-smaller* configuration, such instances of adjunction do not facilitate the MAIF-compliant subextraction of particles. If embedded FocPs′ or ComPs′ do not leave the VP until the matrix clause reaches their sizes, particle subextraction is impossible because the matrix Spec,PartP does not c-command the embedded clause. The impossibility of particle subextraction in such cases is demonstrated with an embedded FocP′ in (99). Finally, long particle movement to Spec,PartP cannot proceed if another particle occurs in the embedded Spec,PartP or the embedded clause is a tensed PartP because no Spec,PartP projects above the PartP-PartP′ adjunction.

(98)



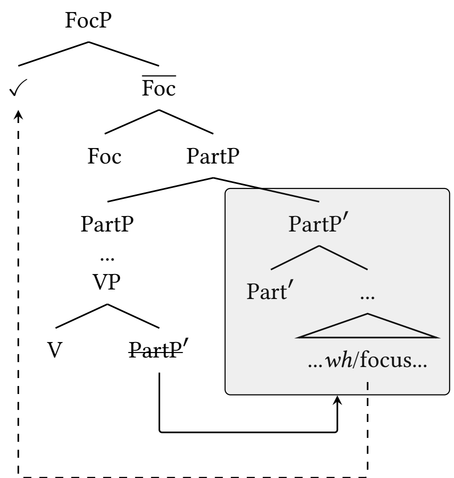
(99)



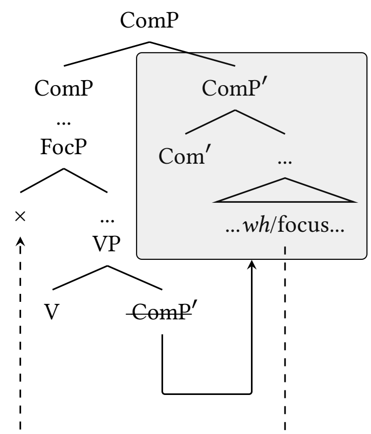
If the embedded clause is a PartP or smaller, subextraction to matrix Spec,Foc is possible. For example, after PartP′

adjoins to matrix PartP in (100), long focus movement proceeds along a MAIF-compliant sequence of heads: V′-sMod′-Part′-Foc. The analysis is similar for smaller embedded clauses like sModPs except that these must be attached lower in order to be transparent for focusing. On the other hand, focus extraction is impossible from ComPs and RelPs. In the system presented here, this is predicted because in order to be open for MAIF-compliant movement, embedded ComPs and RelPs need to adjoin above FocP as illustrated with a ComP in (101). Therefore, no Spec,FocP c-commands both clauses. Again, if an embedded ComP′ adjoins to matrix PartP or lower, we are dealing with a *larger-to-smaller* configuration, from which subextraction violates MAIF.

(100)

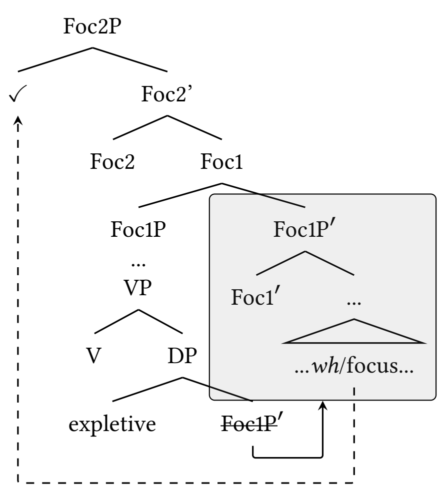


(101)



In bicalusal structures in which a focus/*wh*-constituent moves across another, the embedded Foc1P′ adjoins to the matrix Foc1P in a way that the matrix Spec,Foc2P c-commands Foc1P′, as in (102). In this way, XPs subextracted after the adjunction of Foc1P′ will move along the V′-sMod′-Part′-Foc1′-Foc2 sequence, which is MAIF-compliant.[20]

(102)



[20]As I pointed out in Footnote 3, the number of focus projections is not limited to two in Hungarian, i.e. recursion can in principle bring about a Foc3P, a Foc4P etc. In order to deal with this problem, we could introduce multiple tracks of *fseq*-indices. The main track would track the category labels sMod, Part, Foc, Com, Rel etc.; the secondary track would track sub-indicies e.g. Foc1, Foc2, Foc3. In this way, a well-formed *fseq* would tolerate adjancent heads with the same main index if the sub-index increases. In this way, sMod-Part-Foc-Com; sMod-Part-Foc1-Foc2-Com; sMod-Part-Foc1-Foc2-Foc3-Com would be well-formed sequences but *sMod-Part-**Foc1-Foc1**-Com, *sMod-Part-**Foc3-Foc2**-Com, and *sMod-Part-**Foc1-Foc3**-Com would be bad sequences. Thus, even if movement from an embedded Foc2P cannot proceed to Spec,Foc2P (because no specifier projects above adjuncts), movement to Spec,Foc3P can. Since I do not discuss clauses with more than two foci in the main text, I set this problem aside.

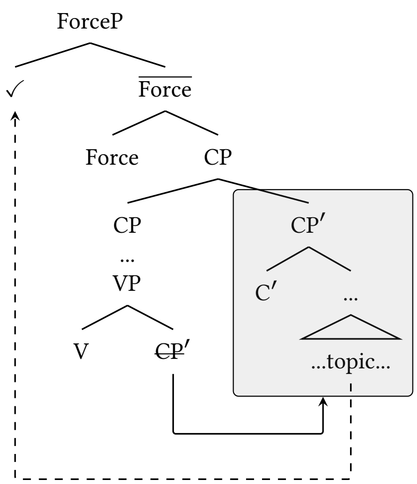
Space does not allow me to derive/re-analyze all cross-linguistic instances of successive cyclicity and Williams Cycle effects/selective opacity, so here I simply sketch the implications of the MAIF-based theory. Another instance of Williams Cycle effects which can receive an explanation from clausal movement comes from German. As discussed in Section 2.3, the opacity pattern of German CPs follows the Strict Williams Cycle in (94): Given that the German *fseq* is V<*v*<T<C<Force, scrambling to Spec,TP in (40) and relativization to Spec,CP in (42) cannot proceed from CPs, whereas topicalization to Spec,ForceP in (41) can. Essentially, similar to Hungarian finite clauses, German CPs are base-generated in a head-final VP(Müller 1999). Unlike Hungarian finite clauses, certain German CPs can be left in their-preverbal position; CPs preceding non-finite verbs can be analyzed as VP-internal (Müller 1999). (Why this violation of the Final-over-Final condition does not lead to ungrammaticality is not discussed here.) However, as shown in (103) (Müller 1999:p.38), topicalization can only proceed from CPs that are moved to the right periphery; in-situ (pre-verbal) CPs are opaque to topicalization. As Müller (2020) discusses, the landing site of CP-extraposition is above TP-adjuncts in the matrix clause, which supports a CP-internal landing-site.

(103) [*Die Maria*]$_1$ *hab-e* *ich* ⟨$_{CP}$ *dass der Fritz* __$_1$ *lieb-t*⟩ *gedacht* ⟨$_{CP}$ *dass der Fritz* __$_1$ *lieb-t*⟩.
the Maria.ACC have-1SG I that the Fritz.NOM love-3SG think.PTCP that the Fritz.NOM love-3SG
'Maria, I thought that Fritz loves.'

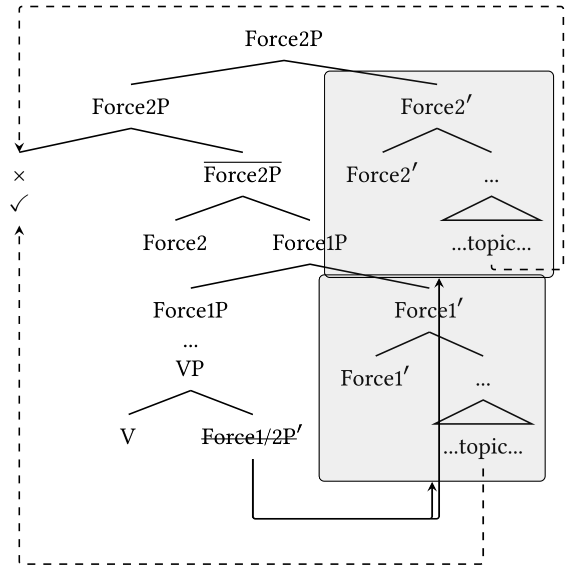
The derivation of (103) is depicted in (104). The embedded CP first moves clause-internally to merge with the matrix CP; this movement proceeds along the MAIF-compliant, clause-internal V-*v*-T-C sequence. After this, the topicalization can proceed from the well-formed V′-*v*′-T′-C′-Force sequence. Assuming that similar to Hungarian finite clauses, German CPs do not have an edge, the MAIF-based theory is explains why in-situ CPs are opaque to sub-extraction: the *V′-*v*′-T′-**C′V**... sequence is not well formed. Furthermore, positions c-commanded by the CP's adjunction site (Spec,TP for scrambling and Spec,CP for relativization) are unavailable from any CPs as the movement of the clause happens too late.[21] One language in which CPs remain in-situ is Tzes. Given that Tzes CPs do not have escapable edges, the lack of CP-movement may be an explanation for the general opacity of Tzes CPs (Polinsky and Potsdam 2001).

[21] As Footnote 9 mentioned, certain German speakers do allow *wh*-movement from CPs to the edge of verb-final clauses, traditionally analyzed as Spec,CP (e.g. Keine 2016, 2020). Whether the landing site of such operations can be re-analyzed as a higher position than unavailable Spec,CP landing site of relativization remains an open question.

(104)



(105)

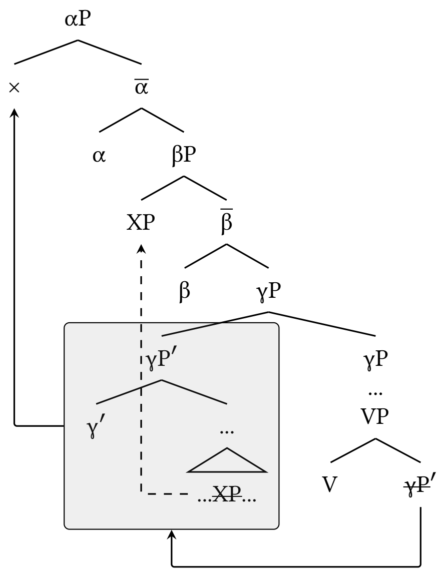


The general implication of the MAIF-based theory is that if a projection in a language allows genuine successive cyclic movement via its edge, its selective opacity cannot receive a MAIF-based explanation. One option in such cases is to provide an independent explanation for selective opacity. Alternatively, reflexes of successive cyclicity (e.g. Georgi 2014, 2017) should receive an independent explanation while keeping the explanation of selective opacity MAIF-based. For example, the German verb-second clauses, which I have been referring to as ForcePs, are opaque to movement to Spec,CP and Spec,TP (Keine 2016, 2020), i.e. they exhibit Williams Cycle effects given that T<C<Force in *fseq*. Keine (2016, 2020) claims that this instance of selective opacity co-occurs with reflexes of successive cyclicity: Any ForceP from which topicalization to another Spec,ForceP has taken place must have an empty Spec,ForceP (i.e. it must be verb-initial) because ForceP has only one specifier if it is occupied by a topic another topic cannot pass through that successive cyclically. The MAIF-based re-analysis of this verb-second/verb-first alternation is depicted in (105): Verb-initial clauses are Force1Ps, verb-second clauses are Force2Ps, where Force1<Force2 in *fseq*. Thus, from a Force1P-adjoined Force1P (i.e. from a verb-initial clause), Spec,Force2P is still accessible. In contrast, Force2P adjoins too high to be transparent to movement to Spec,Force2P.

5.6 Deriving position-dependent opacity

Section 5.4 introduced the idea that in the absence of successive cyclicity, Hungarian γP′ clauses become transparent to the movement of XP to some Spec,βP (such that $\gamma < \beta$ in *fseq*) only if γP′ adjions to equal-sized matrix γP projection as in (106). The node emerging from γP′ and γP in (106) is a γP because both of the lower γPs project. I propose that in such γP-γP′ configurations, the higher γP not only inherits the shared category label of the two γPs (as in Chomsky 2013), but also the union of the feature matrices of the two γPs. In other words, γP′ and γP are no longer maximal projections once another γP emerges above them. The constraint in (107) puts a ban on moving projections that are neither minimal nor maximal, which is a standard assumption in the syntactic literature. Apart from minimal projections, the computation and the interfaces in the *Minimalist Program* can only see maximal projections, which do not project further (Chomsky 1995).[22]

(106)



(107) Projections that are neither minimal nor maximal cannot undergo movement.

Since MAIF only permits subextraction from embedded clauses that have adjoined equal-sized matrix projections, gaining transparency comes with the loss of mobility: once a γP′ adjoined to another γP, it cannot subsequently move to Spec,αP in (106) because γP′ is dominated by a maximal γP. As a consequence, no clause can move to a position c-commanding material extracted from it. In other words, the generalization on position-dependent opacity (repeated in (108)) is derived.

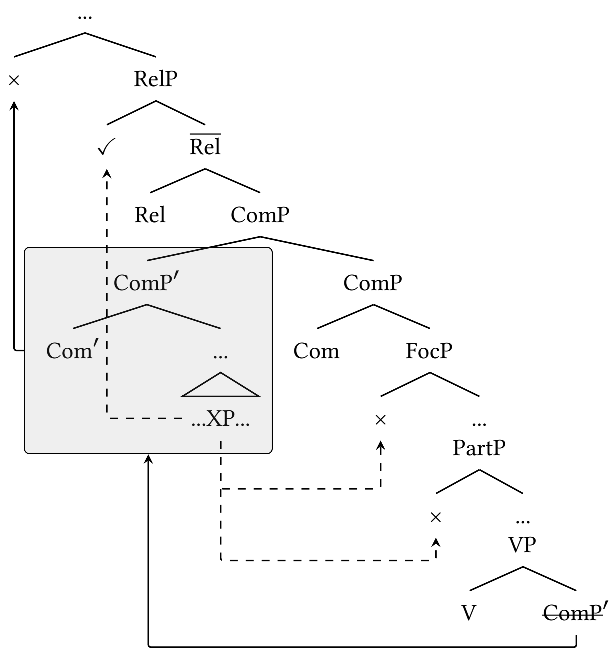
(108) **Position-dependent opacity**: αP cannot move to a position immediately dominated by a βP, if some XP from αP moved to Spec,γP, and α, β, γ ∈ *fseq* F, and β ≥ γ.

This analysis can be applied to the Hungarian data presented in Section 3.3. The diagram in (109) represents a sentence in which an embedded clause ComP′ is moved to the front and gets a topic-interpretation. After moving along the well-formed V-sMod-Part-Foc-Com sequence, ComP′ adjoins to an equal-sized ComP. Since the embedded clause is moved to the left, it is sandwiched between the relative pronouns and foci in the linear order. Thus, this topic position asymmetrically c-commands Spec,PartP and all Spec,FocPs. Consequently, focus and particle subextraction are ruled out from the clause because the ComP′ gains transparency via moving too high/late in the derivation. Crucially, the embedded clause cannot be a smaller clause like sModP′: The sModP-sModP′ adjunction makes the embedded clause immobile by (107) even if the sModP′ is made transparent by this adjunction, i.e. the adjoined sModP′ cannot then be

[22] Possible ways to explain this restriction are *Relativized Minimality* (Rizzi 1990) and the *Minimal Link Condition* (Chomsky 1995): Since a c-commanding probe's search must be minimal, the mother γP is a closer goal than its γP′ daughter as γP has all the features that γP′ has.

topicalized. At the same time, Spec,RelP c-commands the adjoined embedded ComP′ in (109). Therefore, cross-clausal relative movement proceeds along the well-formed V′-sMod′-Part′-Foc′-Com′-Rel sequence, before landing in Spec,RelP. In short, the relative movement is MAIF-compliant. However, ComP′ cannot move further and land in front of the relative pronoun because ComP′ is not a maximal projection as it is dominated by another ComP.

(109)



Topic clauses adjoined to the matrix FocP would have to be FocPs themselves in order allow relative subextraction in the way it was possible in (109). At the same time, since no matrix Spec,PartP and Spec,FocP c-commands such rightward moved FocPs (as topics always adjoin to the outermost focus projection), the subextraction of particles and foci is impossible from FocP-sized topic-clauses as well.

Finally, German CPs can receive a parallel analysis: In a structure like (104), the embedded CP′ is immobile after it has adjoined to the matrix CP because it is dominated by a CP, that is, CPs cannot be topicalized after being CP-adjoined. Only *in-situ* CPs can be topicalized, but *in-situ* CPs are opaque to movement.

5.7 Deriving expletive-mediated position-dependent opacity

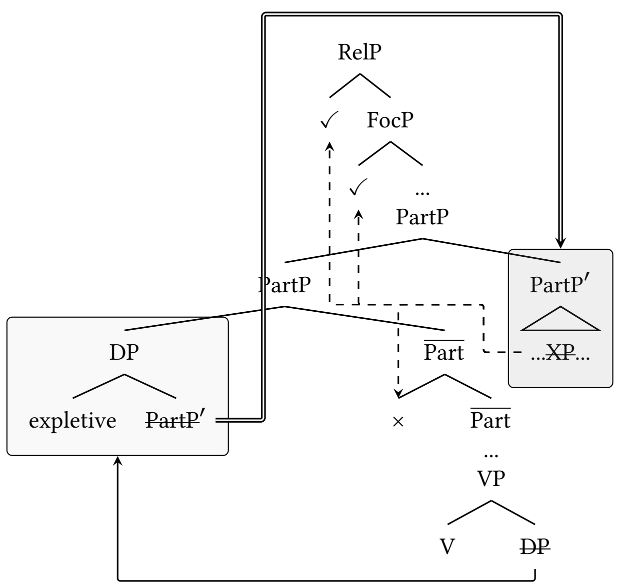
As introduced in Section 3.3, certain Hungarian clauses are generated in a DP headed by an expletive; I assumed that these expletive-headed DPs always leave the matrix VP and land in the left-periphery. Two generalizations were drawn in connection to expletive structures. The first of these is repeated in (110): If an expletive-headed DP moves to a left-periphery specifier, the *hogy*-clause adjoin to the next maximal projection in the matrix clause. That the clause needs to leave the specifier was argued to follow from phonology by Kenesei (1994); that this needs to happen in the next derivational step is derived below from MAIF. The second is repeated in (111): Sub-extraction can only land in a specifier c-commanding the expletive.

(110) If a DP containing an embedded clause introduced by *hogy* occurs in a Spec,αP position in the clausal left-periphery, the clause must adjoin to αP.

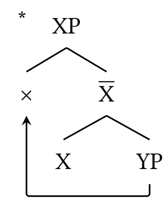
(111) **Expletive-mediated position-dependent opacity:** The expletive associated with αP cannot move to a position immediately dominated by a βP, if some XP from αP moved to Spec,γP, and α, β, γ ∈ *fseq* F, and β ≥ γ.

The tree in (112) depicts the case when the expletive-headed DP moves to Spec,PartP. This first movement step is MAIF-compliant because it happens along the matrix sequence V-sMod-Part. As long as the embedded clause (here a PartP′) is internal to the expletive-headed DP, it is c-commanded by the expletive D head, that is, sub-extraction cannot take place from it in a MAIF-compliant way because sub-extraction would cross the clause-internal heads *and* the expletive D head, which do not belong to the same *fseq*. The clause, here a PartP′, itself has limited possibilities to move: The only position it can reach in a MAIF-complaint way is PartP-adjoined position because it only crosses the expletive D head of the DP. It cannot move to any higher matrix position as it would cross the D head and some matrix head(s) (e.g. Foc, Com, Rel), which, again, do not belong to the same *fseq* as D. Furthermore, the clause is directly the complement of D, that is, irrespectively whether the Hungarian Spec,DP can be used as an escape hatch or not (cf. Szabolcsi 1983), the complement of D cannot move to Spec,DP because this would violate *anti-locality*. Here I am relying on Abels' definition of antilocality in (113) (Abels 2003:p.13), which says that the complement of X cannot move to Spec,XP. In sum, the DP-internal clause is opaque to sub-extraction and the only position it can reach is the PartP-adjoined position. The latter derives the landing site from the generalization in (110): Embedded clauses leaving DPs must adjoin to the next lowest matrix projection.

(112)



(113) *

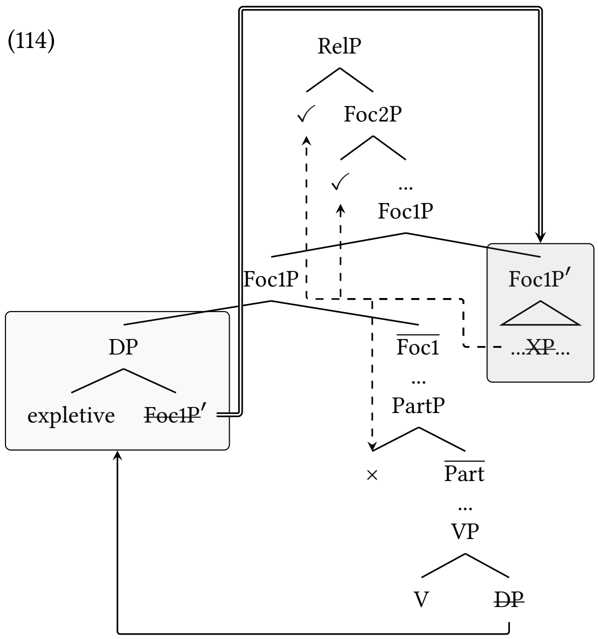


As argued in the previous subsections, the movement of the embedded clause only facilitates subsequent sub-extraction if the embedded clause adjoins to an equal-sized matrix projection. In (112), the embedded clause leaving the DP in the matrix Spec,Part and adjoining to the matrix PartP must be a PartP′ in order to be transparent to *any* movement. Given that the adjoined clause is a PartP, sub-sequent movement can proceed from it to matrix positions like Spec,FocP and Spec,RelP in the same way as it was shown in Section 5.5: Movement along the V′-sMod′-Part′-Foc and V′-sMod′-Part′-Foc-Com-Rel sequences is MAIF-complaint. The embedded PartP′ is an adjunct to the matrix PartP, so no Spec,PartP c-commands the adjunction-site, i.e. particle movement cannot proceed from clauses adjoined to PartP. Since the Spec,PartP position is a criterial position, the expletive-headed DP-remnant cannot undergo subsequent movement to a higher position than either Spec,FocP or Spec,RelP. Furthermore, when the DP is VP-internal, no clause can leave it because at least D and the matrix V would be crossed, which do not belong to the same *fseq*. This derives the generalization in (111): Sub-extraction from clauses can only happen after the expletive-headed DP reaches its criterial position and the clause leaves the DP. The MAIF-based explanation requires the assumption introduced in Section 5.3: in Hungarian, there is no intermediate clausal specifier usable as an intermedaite stop for the expletive-headed DP. This is a necessary assumption because it is essential that the clause cannot be extracted from the DP in a way that the DP can move further.

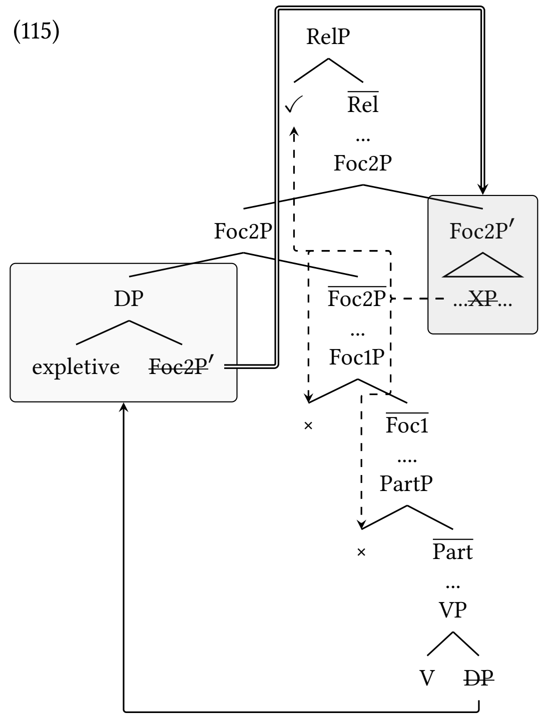
Examples involving expletive-headed DP-remnants in focus positions work in the same fashion. The diagram in (114) depicts a structure in which the expletive-headed DP moves to Spec,Foc1P; the diagram in (115) depicts a structure in which the expletive-headed DP moves to Spec,Foc2P. As argued above, in both cases, no MAIF-compliant sub-extraction is possible from DP-internal clauses, and the clauses themselves can only make one MAIF-compliant movement: adjunction to the next lowest matrix position. (Again, movement in single step to any higher position would proceed along D and one or more clausal heads, and Spec,DP is inaccessible because of antilocality.) Thus, the embedded clause in (114) adjoins to the matrix Foc1P, the embedded clause in (115) adjoins to the matrix Foc2P. In both cases, this movement only crosses the D head, which is MAIF-compliant.

As before, in order to be transparent to *any* sub-extraction, the embedded clause must be a Foc1P in (114) and a Foc2P in (115). The positions c-commanded by the expletives of clauses are inaccessible for sub-extraction from the clauses: That is, Spec,PartP is inaccessible in (114) and Spec,PartP and Spec,Foc1P are inaccessible in (115). The Foc1P-adjoined Foc1P′ in (114) is transparent to sub-extraction to Spec,Foc2P along the well-formed V′-sMod′-Part′-Foc1′-Foc2 sequence and relative sub-extraction along the well-formed V′-sMod′-Part′-Foc1′-Foc2-Com-Rel sequence. The Foc1P-adjoined Foc1P′ in (114) is transparent to sub-extraction to Spec,RelP along the well-formed V′-sMod′-Part′-Foc1′-Foc2′-Com-Rel sequence. Again, since Spec,Foc1P and Spec,Foc2P are criterial positions, the expletive-headed DP-remnant cannot subsequently move from Spec,Foc1P/Spec,Foc2P to a position-commanding the landing site of sub-extraction originating in the clauses of the expletives. That is, (111) is derived for focused expletives, too.

(114)



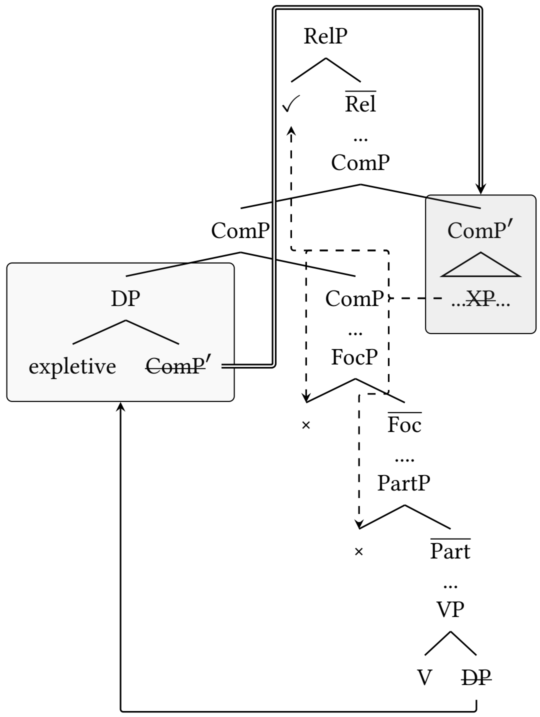
(115)



Finally, the expletive-headed DP-remnants can occur in a topic position, as depicted in (116). As shown in Section 3.3, clauses can but do not have to leave ComP- and FocP-adjoined DPs because (62) does not ban clauses from adjuncts. (116) depicts the case in which the clause leaves the ComP-adjoined expletive-headed DP. If the clause, which is a ComP in (116), does leave the ComP-adjoined DP. The clause can only adjoin to ComP because the movement to RelP would cross Rel and D, which do not belong to the same *fseq*. In contrast, the right-adjunction to the matrix ComP only crosses the D head, i.e. it is MAIF-compliant. As argued in Section 5.5, ComP-adjoined clauses must be ComPs in order to be transparent to *any* sub-extraction. The only specifier c-commanding the ComP-adjoined ComP′ in (116) is Spec,RelP. Movement to this position can proceed along the MAIF-compliant V′-sMod′-Part′-Foc1′-Foc2′-Com′-Rel sequence. In contrast, Spec,PartP, Spec,Foc1P, Spec,Foc2P are c-commanded by the ComP-adjunction site, i.e. they are acessible from clauses whose DP-remnant is ComP-adjoined.

FocP-adjoined topics work in a parallel fashion: Since topics occur above all foci, the only position acessible from a FocP-adjoined topic DP is a FocP-adjoined position, which c-commands the Spec,PartP particle position and all Spec,FocP focus positions. Thus, even if the FocP-adjoined clause is FocP-sized, the only movement that can proceed from it is movement to Spec,RelP along the V′-sMod′-Part′-Foc′-Com-Rel sequence.

(116)



In sum, from clauses contained in an expletive-headed DP, one-step sub-extraction is ruled out by MAIF. After the expletive-headed DP has moved to its criterial position, clauses can only adjoin to matrix projection that hosts the DP-headed expletive. After this adjunction, the clause is transparent to sub-extraction to higher positions iff it is of the same size as the matrix projection serving as the adjunction-site.

6 Conclusion and future directions

The key ingredients for my analysis of the Hungarian size- and position-dependent opacity were the following: (i) syntactic movement is constrained by MAIF, i.e. movement must be internal to extended projections; (ii) embedded clauses are immobile after they adjoin to an equal sized matrix projection; (iii) there is no specifier above the adjunction-site of clauses (optional operations follow obligatory operations); and (iv) there is no successive cyclic movement in Hungarian clause edges.

Due to space reasons, some questions could not be addressed in the current discussion. Firstly, assumption (iv) says that successive cyclicity is in a complementary distribution with size- and position-dependent opacity: The Hungarian clausal domain had size- and position-dependent opacity because no movement could escape them successive cyclically. In order to stay compatible with the MAIF-based theory, projections exhibiting size- or position-dependent opacity as well as reflexes of successive cyclicity need to be re-analyzed. Secondly, my account focused on the derivation of the strict Williams Cycle, which ruled out subextraction from βPs to Spec,βP. It remains an open question how the original Williams Cycle (Williams 1974, 2003, 2013), which permits such movement steps, can be derived by a MAIF-based account.

The relative novelty of the MAIF-based theory lies in that it provides a unification for the standard notions of locality (subjacency, barriers, phases, successive cyclicity etc.) and the various kinds of selective opacity (but also see Müller 2014a,b). The connection between these is not trivial (Keine 2019): While the existing theories of locality (e.g. the Phase Theory) are generally concerned with the longest grammatical movement dependency, the constraints of *size-* and *position-dependent* opacity are restrictions on the relative heights of launching and landing sites across clauses.

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